

The Lambert W Function

Real Branches, Series, Asymptotics, Bounds, Continued Fractions,
Numerical Evaluation, and Applications—with Complete Proofs

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Abstract

The Lambert W function is the inverse, in a multivalued sense, of the map $w \mapsto we^w$. It is the natural function for equations in which an unknown occurs both outside and inside an exponential. This article develops the subject from first principles. Particular attention is paid to the two real branches: the principal branch W_0 and the lower branch W_{-1} . We prove their existence and geometry; derive differential, integral, and algebraic identities; obtain Taylor and power series by Lagrange inversion; develop the common square-root expansion at $-1/e$; derive a unified logarithmic asymptotic expansion for $W_0(x)$ as $x \rightarrow +\infty$ and for $W_{-1}(x)$ as $x \rightarrow 0^-$; prove elementary and sharp branch-specific bounds; and explain integral, continued-fraction, and Schröder-type representations.

The numerical part derives Newton, Halley, logarithmic Newton, and safeguarded iterations, including their error equations and conditioning. Reproducible high-precision experiments compare local series, asymptotic truncations, and global approximations. Applications include exponential-linear equations, $x^x = a$, infinite power towers, delay differential equations, diode circuits, Michaelis–Menten kinetics, Wien-type equations, rooted trees, and asymptotic inversion in prime-number estimates. A final section introduces the complex branches W_k . The exposition assumes only single-variable calculus, basic power series, and elementary complex numbers. The branch-specific assertions and formulas developed here are proved in the text.

Source note. The attached source archive supplied for this article contains the foundational survey of Corless, Gonnet, Hare, Jeffrey, and Knuth, the NIST DLMF entry, papers on guaranteed evaluation, analytic approximations, W_{-1} bounds, continued fractions, Schröder-type series, and applications in circuits, biochemical kinetics, ecology, and prime-number estimates. Those sources are listed in the bibliography. The present text is an independent synthesis: proofs and derivations have been reconstructed, notation has been unified, and principal-branch statements have been extended to W_{-1} whenever the real geometry permits.

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1 The inverse problem behind Lambert W

1.1 Why a new function is needed

The equations

$$3x + 7 = 19, \quad x^2 = 5, \quad e^x = 11$$

are solved by subtraction and division, a square root, and a logarithm, respectively. But consider

$$xe^x = 11. \tag{1.1}$$

Taking a logarithm gives $\log x + x = \log 11$, which still contains x in two different ways. No finite sequence of the usual elementary inverse operations isolates it. The appropriate inverse function is called Lambert W .

Definition 1.1 (Lambert W). A value w of the Lambert W function at z is any solution of

$$we^w = z. \tag{1.2}$$

The complex solutions are organized into branches $W_k(z)$, indexed by integers k . On the real axis, the two branches that can be real are denoted W_0 and W_{-1} .

Thus (1.1) has the solution $x = W_0(11)$. The subscript is not decorative: on part of the negative real axis there are two answers, and the branch must be named.

1.2 The complete real geometry

Set

$$f(w) = we^w, \quad w \in \mathbb{R}.$$

The next theorem contains the entire reason for the two real branches.

Theorem 1.2 (Real branches). *The map $f(w) = we^w$ has the following properties.*

- (i) *It is strictly decreasing on $(-\infty, -1]$ and maps that interval onto $[-1/e, 0)$.*
- (ii) *It is strictly increasing on $[-1, \infty)$ and maps that interval onto $[-1/e, \infty)$.*

Consequently:

- $W_0 : [-1/e, \infty) \rightarrow [-1, \infty)$ is the inverse of f restricted to $[-1, \infty)$ and is strictly increasing;
- $W_{-1} : [-1/e, 0) \rightarrow (-\infty, -1]$ is the inverse of f restricted to $(-\infty, -1]$ and is strictly decreasing;
- for $-1/e < x < 0$, the equation $we^w = x$ has exactly the two real solutions $W_0(x)$ and $W_{-1}(x)$;
- at $x = -1/e$ the two branches meet at the value -1 .

Proof. Differentiate:

$$f'(w) = e^w(1 + w).$$

Because $e^w > 0$, the derivative is negative for $w < -1$, zero at $w = -1$, and positive for $w > -1$. Hence f decreases up to -1 and increases after -1 . Its value at the turning point is

$$f(-1) = -\frac{1}{e}.$$

Moreover,

$$\lim_{w \rightarrow -\infty} we^w = 0^-, \quad \lim_{w \rightarrow +\infty} we^w = +\infty.$$

(The first limit follows, for example, by writing $w = -t$ and using $t/e^t \rightarrow 0$.) The asserted ranges now follow from continuity and strict monotonicity. A continuous strictly monotone function has a continuous strictly monotone inverse on its range, which gives both inverse branches and the solution count. $\square$

Table 1: The two real branches.

Branch	Domain	Range	Monotonicity
W_0	$[-1/e, \infty)$	$[-1, \infty)$	increasing
W_{-1}	$[-1/e, 0)$	$(-\infty, -1]$	decreasing

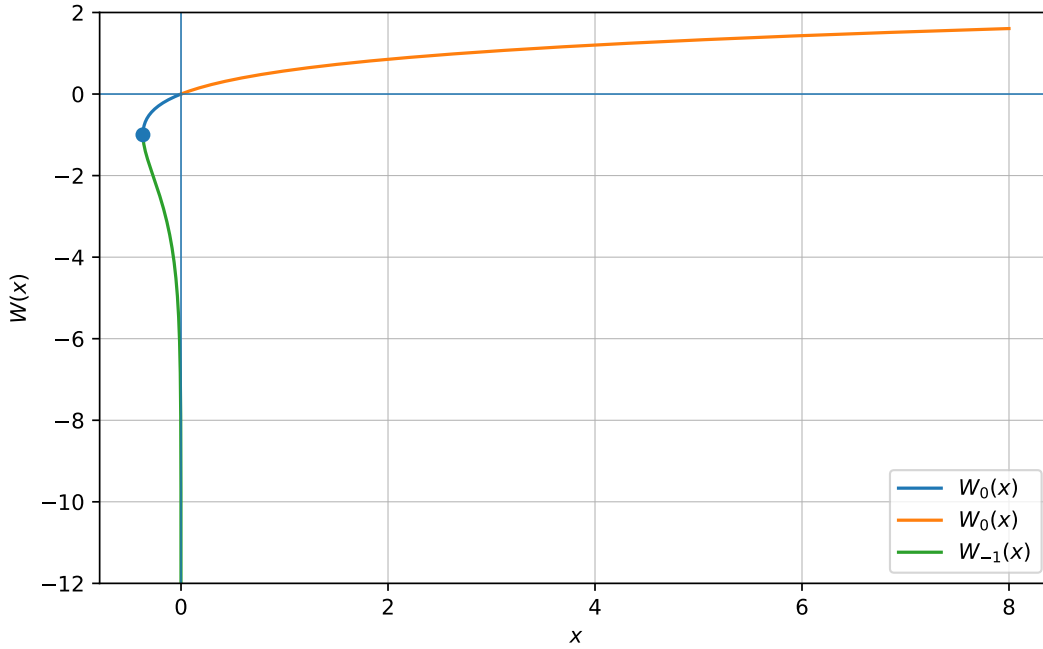


Figure 1: The two real branches. Both pass through $(-1/e, -1)$; only W_0 extends through the origin, while $W_{-1}(x) \rightarrow -\infty$ as $x \rightarrow 0^-$.

1.3 Special values and inverse identities

The definition immediately gives useful exact values:

$$\begin{aligned} W_0(0) &= 0, & W_0(e) &= 1, & W_0(2e^2) &= 2, \\ W_0(-1/e) &= -1, & W_{-1}(-1/e) &= -1, & W_{-1}(-2/e^2) &= -2. \end{aligned} \tag{1.3}$$

More generally, whenever a lies in the range of a specified real branch,

$$W_j(ae^a) = a. \tag{1.4}$$

The phrase “lies in the range” is essential. If $a = -2$, then $ae^a = -2/e^2$ belongs to both real domains, but

$$W_{-1}(-2/e^2) = -2, \quad W_0(-2/e^2) \in (-1, 0).$$

The equation determines a value; the branch determines which value.

A second identity is obtained by dividing (1.2) by W :

$$e^{W_j(z)} = \frac{z}{W_j(z)}, \quad (1.5)$$

provided $z \neq 0$. At $z = 0$ on the principal branch, both sides have the continuous limiting value 1 if the quotient is interpreted by a limit. Taking a real logarithm on either real branch gives

$$W_j(x) + \log |W_j(x)| = \log |x|, \quad (1.6)$$

for $x \neq 0$. This logarithmic form is central to asymptotics and numerical evaluation.

Proposition 1.3 (Limits at the endpoints). *One has*

$$\lim_{x \rightarrow \infty} W_0(x) = \infty, \quad \lim_{x \rightarrow 0^-} W_{-1}(x) = -\infty, \quad \lim_{x \rightarrow 0} W_0(x) = 0.$$

Proof. These are immediate from the ranges and monotonicity in theorem 1.2. For example, if $W_{-1}(x)$ remained bounded below as $x \rightarrow 0^-$, a convergent subsequence would tend to a finite $w \leq -1$ with $we^w = 0$, which is impossible. $\square$

2 Algebraic transformations and branch selection

Lambert W is useful not because every transcendental equation can be forced into it, but because a large and recognizable family can. The target pattern is

$$Ue^U = \text{known quantity}.$$

The safest method is to introduce a new variable U , transform exactly, and only then apply a branch.

2.1 A master linear–exponential equation

Theorem 2.1 (Linear factor times exponential). *Let $a, c \neq 0$. Every solution of*

$$(ax + b)e^{cx+d} = r \quad (2.1)$$

that belongs to a branch W_k is

$$x = \frac{1}{c} W_k \left(\frac{cr}{a} e^{-d+cb/a} \right) - \frac{b}{a}. \quad (2.2)$$

Conversely, every value in (2.2) for which the branch is defined solves (2.1).

Proof. Put $y = x + b/a$, so that $ax + b = ay$ and $cx + d = cy - cb/a + d$. Then (2.1) is equivalent to

$$aye^{cy-cb/a+d} = r,$$

or

$$(cy)e^{cy} = \frac{cr}{a} e^{-d+cb/a}.$$

By the definition of W_k ,

$$cy = W_k \left(\frac{cr}{a} e^{-d+cb/a} \right).$$

Solving for $x = y - b/a$ gives (2.2). Every algebraic step was reversible, proving the converse. $\square$

Example 2.2. Solve $(2x - 3)e^{5x} = 7$. Here $a = 2$, $b = -3$, $c = 5$, $d = 0$, $r = 7$, so

$$x = \frac{1}{5}W_k\left(\frac{35}{2}e^{-15/2}\right) + \frac{3}{2}.$$

The argument is positive, hence only W_0 is real and there is exactly one real solution.

Example 2.3 (Two real answers). Consider $(x + 2)e^x = -1/10$. Formula (2.2) gives

$$x = W_k(-e^2/10) - 2.$$

Since $-e^2/10 < -1/e$, there is no real solution. If the right-hand side is changed to $-1/100$, then the argument $-e^2/100$ lies in $(-1/e, 0)$, and both $k = 0$ and $k = -1$ give real solutions. The branch count is therefore a geometric fact, not a formal choice made after the calculation.

2.2 Useful equivalent forms

The equation $y \log y = A$ is converted by putting $u = \log y$:

$$ue^u = A, \quad y = e^{W_k(A)} = \frac{A}{W_k(A)}.$$

Similarly, if

$$\log y + \alpha y = \beta,$$

then exponentiation yields $ye^{\alpha y} = e^\beta$, and therefore

$$y = \frac{1}{\alpha}W_k(\alpha e^\beta) \tag{2.3}$$

when $\alpha \neq 0$.

Proposition 2.4 (A verification principle). *Let R be a proposed real formula for a branch value at x . Then $R = W_0(x)$ if and only if*

$$Re^R = x \quad \text{and} \quad R \geq -1,$$

and $R = W_{-1}(x)$ if and only if

$$Re^R = x \quad \text{and} \quad R \leq -1.$$

At $x = -1/e$ the two tests agree.

Proof. Necessity is the defining equation together with the branch ranges in theorem 1.2. Conversely, on each stated range the map $R \mapsto Re^R$ is one-to-one. Hence a value satisfying the equation in the correct range must equal the corresponding inverse branch. $\square$

Remark 2.5. Lambert W does not obey the naive scaling rules of a logarithm. The safest way to test any proposed identity is exactly the preceding proposition: call the proposed right-hand side R , compute Re^R , and check the branch range.

3 Differential and integral calculus

Throughout this section, W means any differentiable branch at the point in question. The same calculations therefore apply simultaneously to W_0 and W_{-1} on the interiors of their real domains.

3.1 The first derivative and its differential equation

Theorem 3.1 (Derivative). *If $W(x) \neq -1$ and $W(x)e^{W(x)} = x$, then*

$$W'(x) = \frac{e^{-W(x)}}{1+W(x)} = \frac{W(x)}{x(1+W(x))}. \quad (3.1)$$

The second form is used only when $x \neq 0$. On the principal branch, $W'_0(0) = 1$ by continuity.

Proof. Differentiate $We^W = x$ implicitly:

$$W'e^W + We^W W' = 1.$$

Factoring gives $W'e^W(1+W) = 1$, which proves the first expression. Since $x = We^W$, replacing e^{-W} by W/x proves the second. Finally, $W_0(x)/x = e^{-W_0(x)} \rightarrow 1$ as $x \rightarrow 0$, and $1+W_0(x) \rightarrow 1$, so (3.1) tends to 1. $\square$

Equivalently, every differentiable branch satisfies the nonlinear differential equation

$$x(1+W)W' = W. \quad (3.2)$$

It is singular at the real branch point because $1+W=0$ there.

3.2 Higher derivatives

Repeated differentiation has a clean polynomial structure.

Theorem 3.2 (All higher derivatives). *Define polynomials P_n by*

$$P_1(w) = 1, \quad P_{n+1}(w) = (1+w)P'_n(w) - (nw + 3n - 1)P_n(w). \quad (3.3)$$

Then, for $n \geq 1$,

$$W^{(n)}(x) = e^{-nW(x)} \frac{P_n(W(x))}{(1+W(x))^{2n-1}}. \quad (3.4)$$

Proof. The formula for $n=1$ is (3.1). Suppose it holds for n . Since

$$\frac{d}{dx} = W'(x) \frac{d}{dw} = \frac{e^{-w}}{1+w} \frac{d}{dw} \quad \text{when } w = W(x),$$

we obtain

$$\begin{aligned} W^{(n+1)}(x) &= \frac{e^{-w}}{1+w} \frac{d}{dw} \left(e^{-nw} \frac{P_n(w)}{(1+w)^{2n-1}} \right) \\ &= e^{-(n+1)w} \frac{(1+w)P'_n(w) - (nw + 3n - 1)P_n(w)}{(1+w)^{2n+1}}. \end{aligned}$$

This is exactly (3.4) with n replaced by $n+1$ and P_{n+1} defined by (3.3). Induction completes the proof. $\square$

The first few polynomials are

$$\begin{aligned} P_1(w) &= 1, \\ P_2(w) &= -(w+2), \\ P_3(w) &= 2w^2 + 8w + 9, \\ P_4(w) &= -(6w^3 + 36w^2 + 79w + 64). \end{aligned}$$

The recurrence is often more useful than an expanded closed form, because it allows exact derivatives of arbitrary order to be generated symbolically.

3.3 Monotonicity, concavity, and the inflection of W_{-1}

Putting $P_2(w) = -(w + 2)$ into (3.4) gives

$$W''(x) = -e^{-2W(x)} \frac{W(x) + 2}{(W(x) + 1)^3}. \quad (3.5)$$

Theorem 3.3 (Shape of the real branches). (i) W_0 is strictly increasing and strictly concave on $(-1/e, \infty)$.

(ii) W_{-1} is strictly decreasing on $(-1/e, 0)$. It is strictly convex on $(-1/e, -2/e^2)$, has an inflection point at $x = -2/e^2$, and is strictly concave on $(-2/e^2, 0)$.

Proof. For W_0 , one has $W_0 > -1$, so (3.1) is positive. Also $W_0 + 2 > 0$ and $(W_0 + 1)^3 > 0$, so (3.5) is negative.

For W_{-1} , one has $W_{-1} < -1$, so its first derivative is negative. In (3.5), the denominator $(W_{-1} + 1)^3$ is negative. Therefore the sign of W''_{-1} is the sign of $W_{-1} + 2$. It is positive when $-2 < W_{-1} < -1$, zero at $W_{-1} = -2$, and negative when $W_{-1} < -2$. Because W_{-1} is decreasing and $(-2)e^{-2} = -2/e^2$, these value intervals correspond to the stated x -intervals. $\square$

3.4 Conditioning

Suppose x is perturbed by a small relative amount. The relative condition number of a nonzero value $W(x)$ is

$$\kappa_{\text{rel}}(x) = \left| \frac{xW'(x)}{W(x)} \right| = \frac{1}{|1 + W(x)|}. \quad (3.6)$$

This formula follows directly from (3.1). It explains three numerical facts at once:

- both real branches are ill-conditioned at $x = -1/e$, where $W = -1$;
- W_0 becomes relatively better conditioned as $x \rightarrow \infty$;
- W_{-1} becomes relatively better conditioned as $x \rightarrow 0^-$, even though its absolute magnitude diverges.

The blow-up at $-1/e$ is not a defect of an algorithm. It belongs to the inverse problem itself.

3.5 Antiderivatives

The substitution $x = we^w$ is remarkably effective because

$$dx = e^w(1 + w) dw. \quad (3.7)$$

Theorem 3.4 (Basic antiderivatives). *On any interval belonging to one differentiable branch,*

$$\int W(x) dx = e^{W(x)}(W(x)^2 - W(x) + 1) + C \quad (3.8)$$

$$= xW(x) - x + e^{W(x)} + C, \quad (3.9)$$

$$\int \frac{W(x)}{x} dx = W(x) + \frac{1}{2}W(x)^2 + C, \quad (3.10)$$

$$\int \frac{dx}{1 + W(x)} = e^{W(x)} + C. \quad (3.11)$$

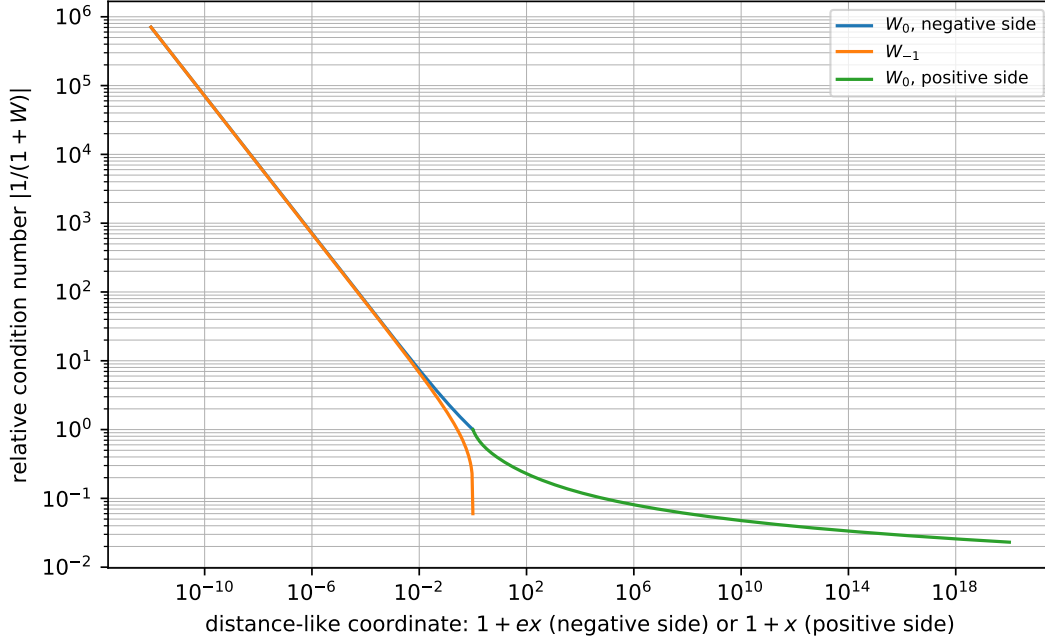


Figure 2: The relative condition number $1/|1+W|$. The square-root branch point is intrinsically ill-conditioned on both real branches.

More generally, if $r \neq 0, -1$ and a consistent real or complex power is used,

$$\int \frac{W(x)^r}{x} dx = \frac{W(x)^r}{r} + \frac{W(x)^{r+1}}{r+1} + C. \quad (3.12)$$

Proof. Set $w = W(x)$ and use (3.7). Then

$$\int W(x) dx = \int w(1+w)e^w dw = e^w(w^2 - w + 1) + C,$$

because differentiating the last expression gives $e^w(w^2 + w)$. Since $x = we^w$, this is also $xw - x + e^w + C$.

Next,

$$\frac{dx}{x} = \frac{1+w}{w} dw.$$

Multiplication by w gives (3.10); multiplication by w^r gives

$$\int (w^{r-1} + w^r) dw,$$

which proves (3.12). Finally, $dx/(1+w) = e^w dw$, proving (3.11). $\square$

Corollary 3.5 (Some exact definite integrals).

$$\int_{-1/e}^0 W_0(x) dx = 1 - \frac{3}{e}, \quad (3.13)$$

$$\int_{-1/e}^0 W_{-1}(x) dx = -\frac{3}{e}, \quad (3.14)$$

$$\int_0^e W_0(x) dx = e - 1. \quad (3.15)$$

The second integral is improper at 0 but converges.

Proof. Use the antiderivative $A(w) = e^w(w^2 - w + 1)$. On W_0 , the endpoint values are $w = -1, 0, 1$ for $x = -1/e, 0, e$. Thus $A(-1) = 3/e$, $A(0) = 1$, and $A(1) = e$, proving the first and third formulas. On W_{-1} , $w \rightarrow -\infty$ as $x \rightarrow 0^-$, and $e^w(w^2 - w + 1) \rightarrow 0$. Subtracting $A(-1) = 3/e$ proves the second formula. $\square$

Remark 3.6. The convergence of (3.14) may look surprising because $W_{-1}(x) \rightarrow -\infty$. Its divergence is only logarithmic; the interval shrinks fast enough for the area to remain finite. The substitution $x = we^w$ makes this transparent.

4 Power series at the origin

Only the principal branch passes through the origin, so an ordinary Taylor series at $x = 0$ belongs to W_0 . The lower branch has no finite value there; its local description near 0^- is logarithmic and will be developed in section 6.

4.1 Lagrange inversion

The coefficient engine is the following form of the Lagrange inversion theorem. We state it for analytic functions, although the proof can also be read purely as an identity of formal power series.

Theorem 4.1 (Lagrange inversion). *Let ϕ be analytic near 0 with $\phi(0) \neq 0$, and let $w = w(x)$ be the unique analytic solution near 0 of*

$$w = x\phi(w), \quad w(0) = 0. \quad (4.1)$$

If F is analytic near 0, then for every $n \geq 1$,

$$[x^n]F(w(x)) = \frac{1}{n}[t^{n-1}]F'(t)\phi(t)^n. \quad (4.2)$$

Here $[x^n]$ means “the coefficient of x^n .”

Proof. We use the residue notation

$$[x^n]G(x) = \operatorname{Res}_{x=0} \frac{G(x)}{x^{n+1}} dx.$$

This is simply Cauchy’s coefficient formula. In (4.1), write $x = t/\phi(t)$. Then

$$dx = \frac{\phi(t) - t\phi'(t)}{\phi(t)^2} dt.$$

Changing variables in the residue gives

$$\begin{aligned} [x^n]F(w(x)) &= \operatorname{Res}_{t=0} F(t) \frac{\phi(t)^{n+1}}{t^{n+1}} \frac{\phi(t) - t\phi'(t)}{\phi(t)^2} dt \\ &= \operatorname{Res}_{t=0} \left(\frac{F(t)\phi(t)^n}{t^{n+1}} - \frac{F(t)\phi(t)^{n-1}\phi'(t)}{t^n} \right) dt. \end{aligned} \quad (4.3)$$

The residue of a derivative is zero. Apply this to $F(t)\phi(t)^n/t^n$:

$$0 = \operatorname{Res}_{t=0} \left(\frac{F'(t)\phi(t)^n}{t^n} + n \frac{F(t)\phi(t)^{n-1}\phi'(t)}{t^n} - n \frac{F(t)\phi(t)^n}{t^{n+1}} \right) dt.$$

Rearranging shows that the residue in (4.3) equals

$$\frac{1}{n} \operatorname{Res}_{t=0} \frac{F'(t)\phi(t)^n}{t^n} dt = \frac{1}{n} [t^{n-1}] F'(t)\phi(t)^n.$$

This proves (4.2). □

4.2 The Taylor series of W_0

The defining equation can be written

$$W_0(x) = xe^{-W_0(x)}, \quad (4.4)$$

so $\phi(t) = e^{-t}$. Taking $F(t) = t$ in theorem 4.1 gives the classical series.

Theorem 4.2 (Taylor series). *For $|x| < 1/e$,*

$$W_0(x) = \sum_{n=1}^{\infty} \frac{(-n)^{n-1}}{n!} x^n. \quad (4.5)$$

In particular,

$$W_0(x) = x - x^2 + \frac{3}{2}x^3 - \frac{8}{3}x^4 + \frac{125}{24}x^5 - \frac{54}{5}x^6 + \frac{16807}{720}x^7 - \dots. \quad (4.6)$$

The radius of convergence is exactly $1/e$.

Proof. By theorem 4.1 and (4.4),

$$[x^n]W_0(x) = \frac{1}{n} [t^{n-1}] e^{-nt} = \frac{1}{n} \frac{(-n)^{n-1}}{(n-1)!} = \frac{(-n)^{n-1}}{n!}.$$

This proves the formal series and hence the analytic expansion in some neighborhood of 0.

To find its radius, let $a_n = (-n)^{n-1}/n!$. Stirling's formula gives

$$|a_n| = \frac{n^{n-1}}{n!} \sim \frac{e^n}{\sqrt{2\pi} n^{3/2}}.$$

Therefore $\lim |a_n|^{1/n} = e$, and the root test gives radius $1/e$. This is also geometrically natural: the nearest singularity of the inverse branch is the critical value $-1/e$. □

Remark 4.3 (What happens on the boundary). At both $x = 1/e$ and $x = -1/e$, the absolute value of the n th term is asymptotic to a constant times $n^{-3/2}$. Hence the series converges absolutely at both endpoints. At $-1/e$ it sums to -1 but cannot be continued as an ordinary power series through that point, because the local behavior is a square root rather than an integer power series.

4.3 Powers and a generalized binomial identity

Lagrange inversion gives much more than W_0 itself.

Theorem 4.4 (Integer powers). *For a positive integer r and $|x| < 1/e$,*

$$W_0(x)^r = \sum_{n=r}^{\infty} \frac{r}{n} \frac{(-n)^{n-r}}{(n-r)!} x^n. \quad (4.7)$$

Proof. Apply theorem 4.1 with $F(t) = t^r$. For $n \geq r$,

$$[x^n]W_0(x)^r = \frac{1}{n}[t^{n-1}]rt^{r-1}e^{-nt} = \frac{r}{n}[t^{n-r}]e^{-nt} = \frac{r}{n} \frac{(-n)^{n-r}}{(n-r)!}.$$

For $n < r$ the coefficient is zero, since $W_0(x)^r$ begins with x^r . □

A useful complex-parameter version is especially compact.

Theorem 4.5 (Powers of $W_0(x)/x$). *For any complex constant α and $|x| < 1/e$,*

$$\left(\frac{W_0(x)}{x}\right)^\alpha = e^{-\alpha W_0(x)} = \sum_{n=0}^{\infty} \frac{\alpha(n+\alpha)^{n-1}}{n!} (-x)^n, \quad (4.8)$$

where the $n = 0$ term is interpreted as 1.

Proof. The first equality follows from $W_0(x)/x = e^{-W_0(x)}$. For $n \geq 1$, use theorem 4.1 with $F(t) = e^{-\alpha t}$:

$$\begin{aligned} [x^n]e^{-\alpha W_0(x)} &= \frac{1}{n}[t^{n-1}](-\alpha)e^{-(n+\alpha)t} \\ &= -\frac{\alpha}{n} \frac{(-(n+\alpha))^{n-1}}{(n-1)!} = \frac{\alpha(n+\alpha)^{n-1}}{n!} (-1)^n. \end{aligned}$$

The constant term is $F(0) = 1$. □

4.4 The tree function and rooted labeled trees

Define

$$T(x) = -W_0(-x). \quad (4.9)$$

Then

$$T(x) = xe^{T(x)}, \quad (4.10)$$

and (4.5) becomes

$$T(x) = \sum_{n=1}^{\infty} \frac{n^{n-1}}{n!} x^n. \quad (4.11)$$

Theorem 4.6 (Cayley, rooted form). *The number of rooted trees whose vertices carry the distinct labels $1, \dots, n$ is n^{n-1} .*

Proof. For $n \geq 2$, the Prüfer code gives a bijection between unrooted labeled trees and sequences of length $n - 2$ with entries in $\{1, \dots, n\}$. To encode a tree, repeatedly remove its smallest-labeled leaf and record the label of that leaf's unique neighbor. After $n - 2$ removals, two vertices remain, and the recorded sequence has length $n - 2$.

To decode a sequence, give each label initial degree one plus the number of times it occurs in the sequence. Read the sequence from left to right. At each step, join the current sequence entry to the smallest label whose current degree is one, then reduce both relevant degrees by one and delete the current entry. At the end, join the two labels of degree one. This produces a tree, and the degree bookkeeping shows that encoding and decoding undo one another. Thus there are n^{n-2} unrooted labeled trees. Choosing one of the n vertices as root gives $n \cdot n^{n-2} = n^{n-1}$ rooted labeled trees. For $n = 1$ there is one tree, agreeing with $1^0 = 1$. □

Thus Lambert W is not merely a device for transcendental equations; its Taylor coefficients encode one of the basic families of combinatorial objects. The functional equation also has a direct combinatorial interpretation. A rooted labeled tree consists of one distinguished root attached to an unordered set of rooted labeled trees. In exponential generating functions, a labeled atom contributes x and an unordered labeled set contributes an exponential, giving $T = xe^T$. Lagrange inversion then recovers the coefficient $n^{n-1}/n!$.

4.5 A local rational approximation

Power series are accurate near 0, but a rational function often reaches farther with the same number of coefficients. Match

$$R(x) = \frac{x(1+ax)}{1+bx+cx^2}$$

to (4.6) through x^4 .

Proposition 4.7 (A Padé approximant of type $(2, 2)$). *The unique rational function of the displayed form whose Taylor expansion agrees with $W_0(x)$ through degree four is*

$$R_{2,2}(x) = \frac{x(1 + \frac{4}{3}x)}{1 + \frac{7}{3}x + \frac{5}{6}x^2}. \quad (4.12)$$

Consequently $W_0(x) - R_{2,2}(x) = \mathcal{O}(x^5)$ as $x \rightarrow 0$.

Proof. The condition

$$(1+bx+cx^2)W_0(x) - x(1+ax) = \mathcal{O}(x^5)$$

and (4.6) give, from the coefficients of x^2, x^3, x^4 ,

$$-1+b-a=0, \quad \frac{3}{2}-b+c=0, \quad -\frac{8}{3}+\frac{3}{2}b-c=0.$$

Solving yields $b = 7/3$, $a = 4/3$, and $c = 5/6$. □

The approximant is not a substitute for branch-aware algorithms, especially near $-1/e$, but it is a useful hand calculation on a moderate interval around zero. Its sampled error is reported in table 2.

5 The square-root branch point at $-1/e$

The two real branches meet where the derivative of we^w vanishes. An ordinary inverse-function Taylor series cannot exist there. The correct local coordinate is a square root, and—importantly—one signed formula describes both W_0 and W_{-1} .

5.1 Finding the natural local variable

Write

$$W(x) = -1 + u. \quad (5.1)$$

Then

$$ex = (u-1)e^u, \quad 1+ex = 1 + (u-1)e^u. \quad (5.2)$$

Expanding at $u = 0$ gives

$$1 + (u - 1)e^u = \sum_{n=2}^{\infty} \frac{n-1}{n!} u^n = \frac{u^2}{2} + \frac{u^3}{3} + \frac{u^4}{8} + \frac{u^5}{30} + \cdots. \quad (5.3)$$

The missing linear term is the analytic signature of the critical point. Define the signed parameter

$$p = \begin{cases} +\sqrt{2(1+ex)}, & W = W_0, \\ -\sqrt{2(1+ex)}, & W = W_{-1}. \end{cases} \quad (5.4)$$

Near the meeting point, p and u have the same sign and $p = u + \mathcal{O}(u^2)$.

5.2 The complete local expansion

Theorem 5.1 (Common Puiseux expansion). *For x sufficiently close to $-1/e$ on either real branch, define p by (5.4). Then*

$$W(x) = -1 + \sum_{n=1}^{\infty} c_n p^n, \quad (5.5)$$

where

$$c_n = \frac{1}{n} [t^{n-1}] \left(\frac{t^2}{2\{1 + (t-1)e^t\}} \right)^{n/2}. \quad (5.6)$$

The first terms are

$$\begin{aligned} W(x) = & -1 + p - \frac{1}{3}p^2 + \frac{11}{72}p^3 - \frac{43}{540}p^4 + \frac{769}{17280}p^5 - \frac{221}{8505}p^6 \\ & + \frac{680863}{43545600}p^7 - \frac{1963}{204120}p^8 + \mathcal{O}(p^9). \end{aligned} \quad (5.7)$$

The same coefficients are used on both branches; only the sign of p changes.

Proof. Define

$$\psi(u) = \frac{\sqrt{2\{1 + (u-1)e^u\}}}{u},$$

where the square root is chosen to have value 1 at $u = 0$. This apparent quotient singularity is removable: by (5.3), the expression inside the square root equals $u^2(1 + \mathcal{O}(u))/u^2$, so ψ is analytic near 0 and $\psi(0) = 1$. Equations (5.2) and (5.4) become

$$p = u\psi(u). \quad (5.8)$$

Thus $u = p\phi(u)$ with $\phi(u) = 1/\psi(u)$. Lagrange inversion theorem 4.1, applied with $F(u) = u$, gives

$$[p^n]u(p) = \frac{1}{n} [t^{n-1}] \phi(t)^n = \frac{1}{n} [t^{n-1}] \left(\frac{t^2}{2\{1 + (t-1)e^t\}} \right)^{n/2}.$$

Since $W = -1 + u$, this proves (5.5) and (5.6). Expanding the coefficient formula gives the listed rational numbers. The analytic inverse-function theorem applies to (5.8), because its derivative at $u = 0$ is 1, so the series converges in some neighborhood of $p = 0$. $\square$

Remark 5.2 (Why this is called a Puiseux series). A Puiseux series permits fractional powers of the original variable. Since p is proportional to $(x + 1/e)^{1/2}$, odd powers of p are half-integer powers of $x + 1/e$. The series is an ordinary Taylor series in p , but a half-power series in $x + 1/e$.

5.3 Leading behavior and derivative blow-up

Taking only the first term of (5.7) yields the most important local approximation.

Corollary 5.3 (Leading branch-point behavior). *As $x \downarrow -1/e$,*

$$W_0(x) = -1 + \sqrt{2e \left(x + \frac{1}{e}\right)} + \mathcal{O}\left(x + \frac{1}{e}\right), \quad (5.9)$$

$$W_{-1}(x) = -1 - \sqrt{2e \left(x + \frac{1}{e}\right)} + \mathcal{O}\left(x + \frac{1}{e}\right). \quad (5.10)$$

Moreover,

$$W'_0(x) \sim +\sqrt{\frac{e}{2(x + 1/e)}}, \quad (5.11)$$

$$W'_{-1}(x) \sim -\sqrt{\frac{e}{2(x + 1/e)}}. \quad (5.12)$$

Proof. Since $p^2 = 2(1 + ex) = 2e(x + 1/e)$, (5.7) gives $W = -1 + p + \mathcal{O}(p^2)$, proving the first two formulas. Differentiating $p^2 = 2(1 + ex)$ gives $p' = e/p$. Also $dW/dp = 1 + \mathcal{O}(p)$, so $W'(x) = e/p + \mathcal{O}(1)$. Substituting the signed value of p proves (5.12). $\square$

The derivative singularity agrees with the condition number in (3.6). It also explains why a numerical routine should switch to the p -series near $-1/e$: direct inversion asks floating-point arithmetic to resolve a nearly horizontal graph.

5.4 A coefficient recurrence for computation

Formula (5.6) is explicit, but a recurrence is convenient when many coefficients are desired. Let

$$u(p) = \sum_{n \geq 1} c_n p^n.$$

Substituting into

$$\frac{p^2}{2} = 1 + (u - 1)e^u \quad (5.13)$$

and comparing coefficients determines c_n successively. Indeed, the coefficient of p^{N+1} is linear in c_N : the quadratic leading term $u^2/2$ contributes $c_N p^{N+1}$ through $c_1 c_N p^{N+1}$, and $c_1 = 1$; all other contributions involve only $c_1, \dots, c_{N-1}$. Therefore the recurrence is triangular and exact. This is the method used in the accompanying code to check printed coefficients.

Proposition 5.4 (Branch separation). *As $x \downarrow -1/e$,*

$$W_0(x) - W_{-1}(x) = 2\sqrt{2(1 + ex)} + \frac{11}{36}\{2(1 + ex)\}^{3/2} + \mathcal{O}((1 + ex)^{5/2}). \quad (5.14)$$

Proof. Evaluate (5.7) at $p = q$ and $p = -q$, where $q = \sqrt{2(1 + ex)}$. The even powers cancel in the difference and the odd powers double. Keeping p and p^3 gives the stated formula. $\square$

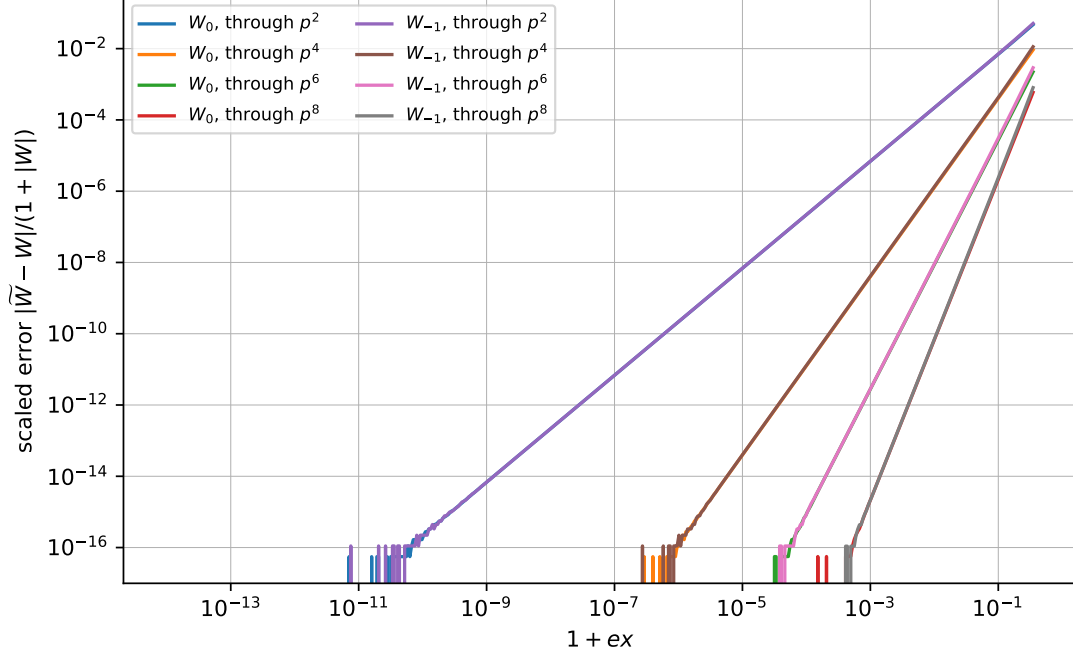


Figure 3: Sampled scaled errors of truncations of the signed- p expansion on both real branches. The common coefficient sequence works on W_0 and W_{-1} ; branch selection is encoded entirely by the sign of p .

6 Logarithmic asymptotic expansions

There are two real limits in which $|W|$ becomes large:

$$W_0(x) \rightarrow +\infty \quad (x \rightarrow +\infty), \quad W_{-1}(x) \rightarrow -\infty \quad (x \rightarrow 0^-).$$

The same expansion governs both. This is one of the most useful principal-branch results that extends almost unchanged to W_{-1} .

6.1 The first two terms

For $x > 0$, (1.6) reads

$$W_0(x) + \log W_0(x) = \log x.$$

The logarithm is smaller than its argument, so a first guess is $W_0(x) \approx \log x$. Feeding this back into the logarithm gives

$$W_0(x) \approx \log x - \log \log x. \quad (6.1)$$

For $W_{-1}(x)$, $x < 0$ and $W_{-1}(x) < 0$, so

$$W_{-1}(x) + \log(-W_{-1}(x)) = \log(-x).$$

As $x \rightarrow 0^-$, the right side tends to $-\infty$, and the analogous first approximation is

$$W_{-1}(x) \approx \log(-x) - \log(-\log(-x)). \quad (6.2)$$

The second logarithm has the same form in both cases once absolute values are inserted.

6.2 A unified coefficient formula

For either of the two limits, define

$$L_1 = \begin{cases} \log x, & W_0(x), \ x \rightarrow +\infty, \\ \log(-x), & W_{-1}(x), \ x \rightarrow 0^-, \end{cases} \quad L_2 = \log |L_1|. \quad (6.3)$$

Thus $L_1 \rightarrow +\infty$ in the first case and $L_1 \rightarrow -\infty$ in the second. In both cases $L_2 > 0$ eventually.

Theorem 6.1 (Unified real asymptotic expansion). *In either real limit described in (6.3), and for every fixed integer $N \geq 0$,*

$$W(x) = L_1 - L_2 + \sum_{j=1}^N \frac{P_j(L_2)}{L_1^j} + \mathcal{O}\left(\frac{L_2^{N+1}}{|L_1|^{N+1}}\right), \quad (6.4)$$

where

$$P_j(L) = \frac{1}{j} [z^{j-1}] \frac{(L + \log(1-z))^j}{1-z}. \quad (6.5)$$

The first four correction polynomials are

$$P_1(L) = L, \quad (6.6)$$

$$P_2(L) = \frac{L(-2+L)}{2}, \quad (6.7)$$

$$P_3(L) = \frac{L(6-9L+2L^2)}{6}, \quad (6.8)$$

$$P_4(L) = \frac{L(-12+36L-22L^2+3L^3)}{12}. \quad (6.9)$$

Consequently,

$$\begin{aligned} W(x) = L_1 - L_2 + \frac{L_2}{L_1} + \frac{L_2(-2+L_2)}{2L_1^2} + \frac{L_2(6-9L_2+2L_2^2)}{6L_1^3} \\ + \frac{L_2(-12+36L_2-22L_2^2+3L_2^3)}{12L_1^4} + \mathcal{O}\left(\frac{L_2^5}{|L_1|^5}\right). \end{aligned} \quad (6.10)$$

Proof. In either real regime, W and L_1 have the same sign. Write

$$W = L_1 - L + v, \quad L = L_2, \quad t = \frac{1}{L_1}. \quad (6.11)$$

Then

$$|W| = |L_1|(1 - tL + tv)$$

for sufficiently large $|L_1|$. Substitution into $W + \log |W| = L_1$ yields

$$v = -\log(1 - tL + tv). \quad (6.12)$$

Set

$$u = t(L - v). \quad (6.13)$$

Then $1 - u = 1 - tL + tv$, so

$$v = -\log(1 - u), \quad u = t(L + \log(1 - u)). \quad (6.14)$$

For the moment regard L as a fixed parameter. Apply Lagrange inversion to $u = t\phi(u)$ with $\phi(u) = L + \log(1 - u)$ and $F(u) = -\log(1 - u)$. Since $F'(u) = 1/(1 - u)$,

$$[t^j]v(t) = \frac{1}{j}[z^{j-1}]\frac{(L + \log(1 - z))^j}{1 - z} = P_j(L).$$

This proves the formal expansion $v = \sum_{j \geq 1} P_j(L)t^j$ and formula (6.5). Direct expansion gives (6.9).

It remains to justify asymptotic use when $L = \log |L_1|$ is not fixed. Let $v_N = \sum_{j=1}^N P_j(L)t^j$ and $w_N = L_1 - L + v_N$. Since $|t|L \rightarrow 0$, the Taylor expansion of the logarithm in (6.12) is valid for all sufficiently large $|L_1|$. The coefficient construction cancels every power through t^N , so

$$w_N + \log |w_N| - L_1 = \mathcal{O}(|t|^{N+1}L^{N+1}).$$

Let $F(w) = w + \log |w| - L_1$. On the relevant large-magnitude interval, $F'(w) = 1 + 1/w$ is bounded away from zero. By the mean value theorem, for some ξ between the exact root W and w_N ,

$$W - w_N = -\frac{F(w_N)}{F'(\xi)}.$$

Combining this with (6.2) proves the remainder in (6.4). $\square$

6.3 Stirling numbers and the double expansion

The unsigned Stirling number of the first kind $\begin{bmatrix} n \\ r \end{bmatrix}$ counts permutations of n objects having exactly r cycles. Algebraically it is defined by

$$y(y+1) \cdots (y+n-1) = \sum_{r=0}^n \begin{bmatrix} n \\ r \end{bmatrix} y^r. \quad (6.15)$$

These numbers give a closed form for every polynomial P_j .

Lemma 6.2 (Stirling form of the coefficient polynomials). *For $j \geq 1$,*

$$P_j(L) = \sum_{m=1}^j \frac{(-1)^{j-m}}{m!} \begin{bmatrix} j \\ j-m+1 \end{bmatrix} L^m. \quad (6.16)$$

Proof. The standard exponential generating function for unsigned Stirling numbers is

$$\frac{(-\log(1-z))^r}{r!} = \sum_{n=r}^{\infty} \begin{bmatrix} n \\ r \end{bmatrix} \frac{z^n}{n!}. \quad (6.17)$$

Differentiate (6.17) with $r = q+1$ to obtain

$$[z^{j-1}] \frac{(\log(1-z))^q}{1-z} = (-1)^q \frac{q!}{(j-1)!} \begin{bmatrix} j \\ q+1 \end{bmatrix}. \quad (6.18)$$

Expand $(L + \log(1-z))^j$ in (6.5). The coefficient of L^m uses $q = j-m$ powers of the logarithm and equals

$$\begin{aligned} \frac{1}{j} \begin{bmatrix} j \\ m \end{bmatrix} [z^{j-1}] \frac{(\log(1-z))^{j-m}}{1-z} &= \frac{1}{j} \begin{bmatrix} j \\ m \end{bmatrix} (-1)^{j-m} \frac{(j-m)!}{(j-1)!} \begin{bmatrix} j \\ j-m+1 \end{bmatrix} \\ &= \frac{(-1)^{j-m}}{m!} \begin{bmatrix} j \\ j-m+1 \end{bmatrix}. \end{aligned}$$

This proves (6.16). $\square$

Putting $j = n + m$ in (6.16) gives the traditional bivariate form.

Corollary 6.3 (Comtet–de Bruijn double series). *Formally, and asymptotically when truncated by total degree,*

$$W = L_1 - L_2 + \sum_{n=0}^{\infty} \sum_{m=1}^{\infty} \frac{(-1)^n}{m!} \begin{bmatrix} n+m \\ n+1 \end{bmatrix} \frac{L_2^m}{L_1^{n+m}}. \quad (6.19)$$

Proof. In (6.16), set $j = n + m$. Then $j - m + 1 = n + 1$, and the term $P_j(L_2)/L_1^j$ becomes exactly the summand in (6.19). Summing over all $j \geq 1$ is equivalent to summing over $n \geq 0$ and $m \geq 1$. $\square$

6.4 How to read the expansion on each branch

For $W_0(x)$ at large positive x ,

$$L_1 = \log x, \quad L_2 = \log \log x.$$

All denominators in (6.10) are positive. For $W_{-1}(x)$ near 0^- ,

$$L_1 = \log(-x) < 0, \quad L_2 = \log(-\log(-x)).$$

Odd powers of L_1 are negative. No separate coefficient table is needed: the signs appropriate to the lower branch are already carried by L_1 .

Example 6.4. Take $x = -10^{-20}$. Then $L_1 = -20 \log 10 \approx -46.0517$ and $L_2 \approx 3.8298$. The first two terms give $L_1 - L_2 \approx -49.8815$; adding L_2/L_1 moves the approximation to about -49.9647 . The exact value is approximately -49.9630 . The lower branch is therefore not approximated by $\log(-x)$ alone: the second logarithm is a visible correction even at very small $|x|$.

6.5 A simple rigorous two-term enclosure for W_0

The first two asymptotic terms also form elementary bounds.

Proposition 6.5 (Logarithmic enclosure). *For $x > e$, put $L_1 = \log x$ and $L_2 = \log L_1$. Then*

$$L_1 - L_2 < W_0(x) < L_1. \quad (6.20)$$

Proof. Here $W_0(x) > 1$. From $W_0 = L_1 - \log W_0$ and $\log W_0 > 0$, one gets $W_0 < L_1$. Since the logarithm is increasing, this implies $\log W_0 < \log L_1 = L_2$. Substitution back into $W_0 = L_1 - \log W_0$ gives $W_0 > L_1 - L_2$. $\square$

The lower endpoint in (6.20) is not merely a bound; it is an excellent starting value for a monotone quadratic iteration developed in section 9.

7 Inequalities on the real branches

Approximations tell us what is likely; inequalities tell us what is guaranteed. This section develops bounds that require only elementary calculus. The principal branch and the lower branch need different natural variables.

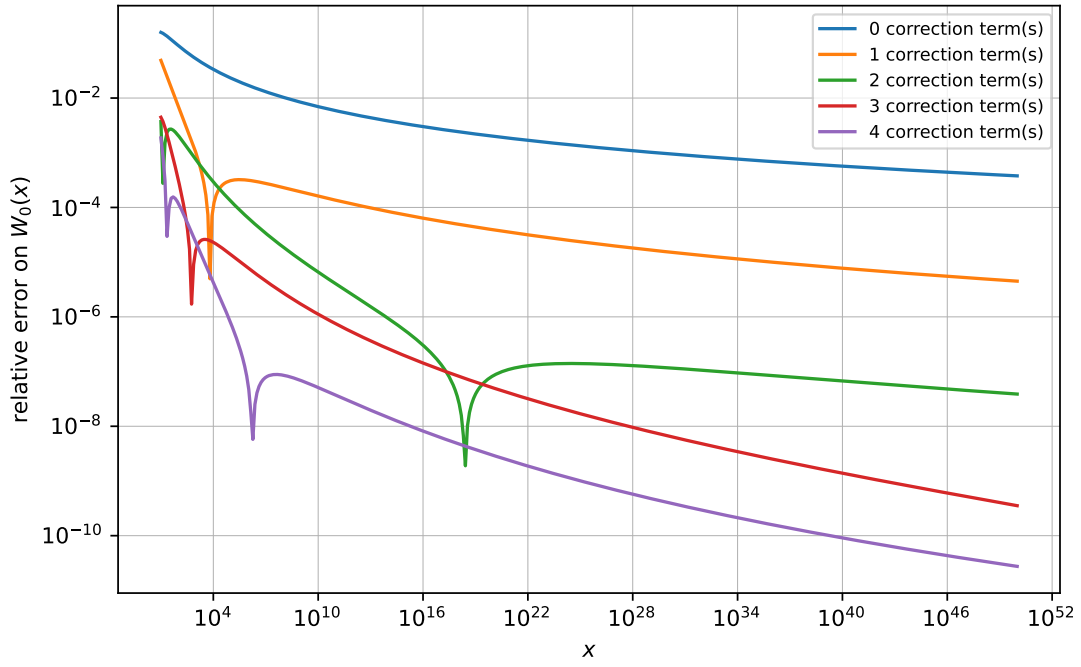


Figure 4: Relative errors of successive truncations of the logarithmic expansion for $W_0(x)$ as x grows. An asymptotic expansion need not improve monotonically at every moderate x , but each fixed truncation has the proved large- x order.

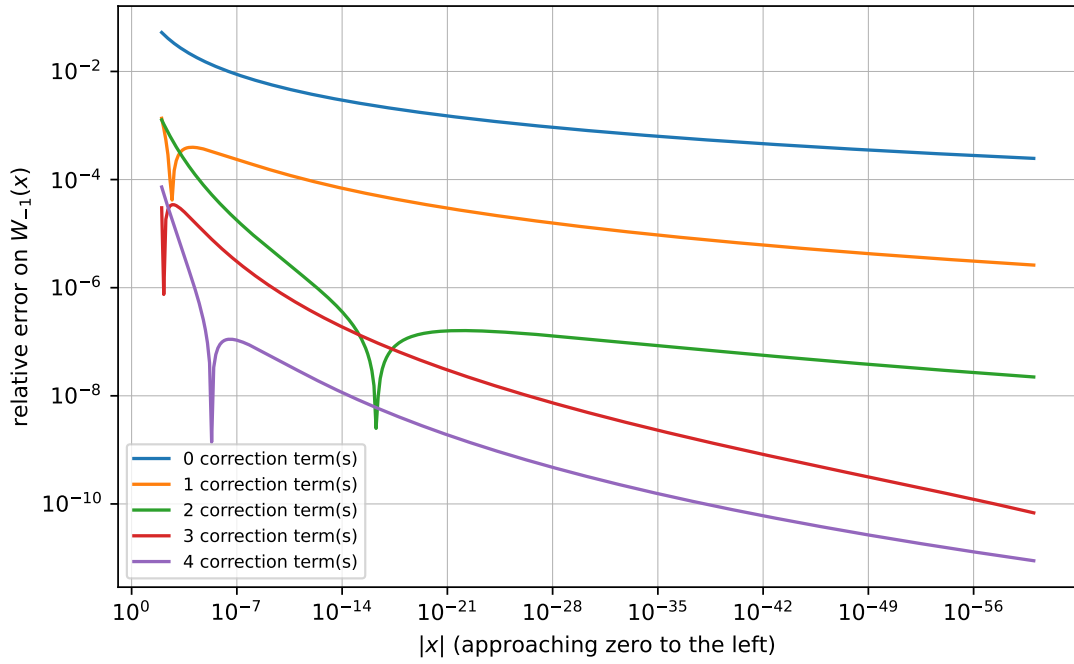


Figure 5: The same coefficient polynomials applied to $W_{-1}(x)$ as $x \rightarrow 0^-$. The horizontal axis is $|x|$ and runs toward zero from left to right.

7.1 Global elementary bounds for W_0

Theorem 7.1 (Three bounds for nonnegative arguments). *For every $x \geq 0$,*

$$\frac{x}{1+x} \leq W_0(x) \leq \log(1+x) \leq x. \quad (7.1)$$

The first two inequalities are strict when $x > 0$ except for limiting equality at 0.

Proof. Put $w = W_0(x) \geq 0$, so $x = we^w$.

First, $W_0(x) \leq x$ follows immediately from $e^w \geq 1$. The familiar inequality $\log(1+x) \leq x$ follows because $x - \log(1+x)$ has derivative $x/(1+x) \geq 0$ and vanishes at 0.

To prove $w \leq \log(1+x)$, it is enough to prove $e^w \leq 1+x$. But

$$1+x - e^w = 1+(w-1)e^w.$$

The right side is zero at $w = 0$ and has derivative $we^w \geq 0$, so it is nonnegative.

Finally, $x/(1+x) \leq w$ is equivalent, for $w > 0$, to

$$e^w \leq 1 + we^w,$$

or $1 \leq w + e^{-w}$. The function $w + e^{-w}$ equals 1 at 0 and has derivative $1 - e^{-w} \geq 0$. Thus the inequality holds. The case $x = 0$ follows directly. $\square$

Corollary 7.2 (A bracket for computation). *For $x > 0$,*

$$\frac{x}{1+x} < W_0(x) < \log(1+x). \quad (7.2)$$

Thus these two elementary expressions provide a finite bracket for bisection or safeguarded Newton iteration on the entire positive half-line.

For $-1/e < x < 0$, the simplest exact bracket is already supplied by the branch range:

$$-1 < W_0(x) < 0. \quad (7.3)$$

Near $-1/e$, the Puiseux expansion gives much narrower local brackets if its remainder is controlled.

7.2 A natural coordinate for W_{-1}

Every $x \in (-1/e, 0)$ can be written uniquely as

$$x = -e^{-u-1}, \quad u = -1 - \log(-x) > 0. \quad (7.4)$$

Write

$$W_{-1}(x) = -1 - y, \quad y > 0. \quad (7.5)$$

Substituting into $We^W = x$ gives

$$-(1+y)e^{-1-y} = -e^{-1-u},$$

so

$$u = y - \log(1+y). \quad (7.6)$$

The function $g(y) = y - \log(1+y)$ is positive and strictly increasing for $y > 0$, because $g'(y) = y/(1+y) > 0$. Bounding its inverse produces bounds for W_{-1} .

7.3 Sharp elementary bounds for W_{-1}

The following inequalities, adapted from Chatzigeorgiou's branch-specific bounds [3], work on the entire domain and have the correct square-root scale at the branch point.

Theorem 7.3 (Global W_{-1} enclosure). *Let $-1/e < x < 0$ and define $u = -1 - \log(-x) > 0$. Then*

$$-1 - \sqrt{2u} - u < W_{-1}(x) < -1 - \sqrt{2u} - \frac{2}{3}u. \quad (7.7)$$

Proof. Let $y = -1 - W_{-1}(x) > 0$. By (7.6), $u = g(y) = y - \log(1 + y)$. We will prove

$$\frac{2}{3}g(y) < y - \sqrt{2g(y)} < g(y). \quad (7.8)$$

Replacing $g(y)$ by u and rearranging then gives

$$\sqrt{2u} + \frac{2}{3}u < y < \sqrt{2u} + u.$$

Since $W_{-1} = -1 - y$, reversing signs produces (7.7).

For the right inequality in (7.8), observe that $y - g(y) = \log(1 + y)$. Put $t = \log(1 + y) > 0$, so $y = e^t - 1$ and $g(y) = e^t - 1 - t$. The desired inequality is equivalent to

$$t < \sqrt{2(e^t - 1 - t)}.$$

After squaring, this follows from

$$e^t > 1 + t + \frac{t^2}{2} \quad (t > 0),$$

which is the exponential series with positive omitted terms.

For the left inequality in (7.8), we need

$$y - \frac{2}{3}g(y) > \sqrt{2g(y)}.$$

The left side is positive and equals $(e^t - 1 + 2t)/3$. Squaring is therefore legitimate. The squared inequality is

$$(e^t - 1 + 2t)^2 > 18(e^t - 1 - t). \quad (7.9)$$

Expanding both exponentials gives the exact series

$$(e^t - 1 + 2t)^2 - 18(e^t - 1 - t) = \sum_{n=4}^{\infty} \frac{2^n + 4n - 20}{n!} t^n. \quad (7.10)$$

For $n \geq 4$, $2^n + 4n - 20 > 0$, so every term is positive. This proves (7.9), hence the left inequality in (7.8), and therefore the theorem. $\square$

Remark 7.4 (Behavior of the enclosure). As $u \rightarrow 0$, both bounds begin with $-1 - \sqrt{2u}$, exactly matching the square-root branch behavior. As $u \rightarrow \infty$, their gap is $u/3$. The gap is not asymptotically small compared with $|W_{-1}| \sim u$, but the bounds remain finite, explicit, and branch-safe for every $x \in (-1/e, 0)$, making them valuable as a guaranteed numerical bracket.

Corollary 7.5 (A monotone lower starter). *With u as above,*

$$\beta_0 = -1 - \sqrt{2u} - u \quad (7.11)$$

obeys $\beta_0 < W_{-1}(x) < -1$. It can therefore initialize the monotone logarithmic Newton iteration proved in theorem 9.5.

Proof. The first inequality is the left half of (7.7); the second is the range of W_{-1} . $\square$

7.4 A comparison principle

Many further bounds can be proved without manipulating W directly.

Proposition 7.6 (Comparison by the defining map). *Let x lie in the domain of a real branch.*

- (i) *On the principal range $[-1, \infty)$, if $a, b \geq -1$ and $ae^a \leq x \leq be^b$, then $a \leq W_0(x) \leq b$.*
- (ii) *On the lower range $(-\infty, -1]$, if $a \leq b \leq -1$ and $ae^a \geq x \geq be^b$, then $a \leq W_{-1}(x) \leq b$.*

Proof. On $[-1, \infty)$ the defining map is increasing; on $(-\infty, -1]$ it is decreasing, by theorem 1.2. Applying the appropriate inverse preserves or reverses inequalities exactly as stated. $\square$

This comparison principle is a practical recipe. To prove a proposed upper bound $B(x)$ for W_0 , it is enough to prove $B(x) \geq -1$ and $B(x)e^{B(x)} \geq x$. For W_{-1} , the monotonicity direction is reversed.

8 Integral, continued-fraction, and inverse-series representations

A special function becomes more useful when it can be represented in several ways. Series are good near a center, continued fractions can reorganize series, integrals connect the function to harmonic analysis, and inverse series generate high-order root corrections.

8.1 A logarithmic integral representation

The following representation is a convenient consequence of the Taylor series. Corcino, Corcino, and Mező extend a closely related formula to a larger real interval; the restricted form below is enough for an elementary, fully justified proof.

Theorem 8.1 (Logarithmic integral). *For real x with $|x| < 1/e$,*

$$W_0(x) = \frac{1}{\pi} \operatorname{Re} \int_0^\pi \log \left(\frac{\exp(e^{it}) - xe^{-it}}{\exp(e^{it}) - xe^{it}} \right) dt. \quad (8.1)$$

The logarithm is the analytic determination obtained from its convergent power series in the proof.

Proof. We first prove the Fourier identity

$$\frac{a^n}{n!} = \frac{2}{\pi} \operatorname{Im} \int_0^\pi \exp(ae^{it}) \sin(nt) dt \quad (8.2)$$

for real a and positive integer n . Expand the exponential uniformly on $[0, \pi]$:

$$\exp(ae^{it}) = \sum_{m=0}^{\infty} \frac{a^m}{m!} e^{imt}.$$

The imaginary part is $\sum_{m \geq 1} a^m \sin(mt)/m!$. Orthogonality gives

$$\int_0^\pi \sin(mt) \sin(nt) dt = \begin{cases} \pi/2, & m = n, \\ 0, & m \neq n, \end{cases}$$

which proves (8.2).

Apply (8.2) with $a = -n$. Since $(-n)^{n-1} = -(-n)^n/n$, the Taylor series (4.5) becomes

$$W_0(x) = -\frac{2}{\pi} \operatorname{Im} \int_0^\pi \sum_{n=1}^\infty \frac{\{x \exp(-e^{it})\}^n}{n} \sin(nt) dt. \quad (8.3)$$

For $|x| < 1/e$,

$$|x \exp(-e^{it})| = |x| \exp(-\cos t) \leq |x|e < 1,$$

so the series converges uniformly and may be integrated term by term. Put $r = x \exp(-e^{it})$. The logarithm series gives

$$\begin{aligned} \sum_{n=1}^\infty \frac{r^n}{n} \sin(nt) &= \frac{1}{2i} \sum_{n=1}^\infty \frac{(re^{it})^n - (re^{-it})^n}{n} \\ &= \frac{1}{2i} \log \left(\frac{1 - re^{-it}}{1 - re^{it}} \right). \end{aligned}$$

For any complex number L , $-2 \operatorname{Im}(L/(2i)) = \operatorname{Re} L$. Therefore (8.3) is

$$\frac{1}{\pi} \operatorname{Re} \int_0^\pi \log \left(\frac{1 - re^{-it}}{1 - re^{it}} \right) dt.$$

Multiplying numerator and denominator inside the logarithm by $\exp(e^{it})$ gives (8.1). $\square$

Remark 8.2. The representation is not intended as the fastest numerical algorithm. Its value is structural: the rooted-tree coefficients of W_0 emerge from Fourier orthogonality, and an inverse function is expressed as an average of a logarithm.

8.2 Euler's continued fraction for a series

We use the generalized continued-fraction notation

$$b_0 + \mathbf{K}_{m=1}^N \frac{a_m}{b_m} = b_0 + \frac{a_1}{b_1 + \frac{a_2}{b_2 + \cdots + \frac{a_N}{b_N}}}.$$

Lemma 8.3 (Euler transformation). *Let $c_1, \dots, c_N$ be nonzero and put $r_m = c_{m+1}/c_m$. Then*

$$\sum_{j=0}^N c_j = c_0 + \frac{c_1}{1 + \frac{-r_1}{1 + r_1 + \frac{-r_2}{1 + r_2 + \cdots + \frac{-r_{N-1}}{1 + r_{N-1}}}}}. \quad (8.4)$$

If $\sum c_j$ converges and the finite continued fractions have a limit, the corresponding infinite fraction equals the series sum.

Proof. For $1 \leq m \leq N-1$, define

$$Q_m = 1 + r_m + r_m r_{m+1} + \cdots + r_m r_{m+1} \cdots r_{N-1}.$$

Then

$$Q_m = 1 + r_m Q_{m+1}, \quad Q_{N-1} = 1 + r_{N-1}. \quad (8.5)$$

A backward induction using (8.5) shows that the denominator beginning at level m in (8.4) equals Q_m/Q_{m+1} , with the natural convention $Q_N = 1$. At the top level,

$$1 - \frac{r_1}{Q_1/Q_2} = 1 - \frac{r_1 Q_2}{Q_1} = \frac{1}{Q_1},$$

by $Q_1 = 1 + r_1 Q_2$. Hence the fraction after c_0 equals $c_1 Q_1$. But

$$c_1 Q_1 = c_1 + c_1 r_1 + c_1 r_1 r_2 + \cdots = c_1 + c_2 + \cdots + c_N.$$

Adding c_0 proves the finite identity. Passing to a limit proves the last statement whenever that limit exists. $\square$

8.3 A continued fraction for W_0

Apply theorem 8.3 to the Taylor terms

$$c_m = \frac{(-m)^{m-1}}{m!} x^m, \quad m \geq 1.$$

Their ratios are

$$\frac{c_m}{c_{m-1}} = - \left(\frac{m}{m-1} \right)^{m-2} x, \quad m \geq 2. \quad (8.6)$$

Define

$$b_m = \left(1 - \frac{1}{m} \right)^{2-m} = \left(\frac{m}{m-1} \right)^{m-2} > 0. \quad (8.7)$$

Theorem 8.4 (Euler continued fraction for W_0). *For $|x| < 1/e$,*

$$W_0(x) = \frac{x}{1 + \frac{b_2 x}{1 - b_2 x + \frac{b_3 x}{1 - b_3 x + \frac{b_4 x}{1 - b_4 x + \cdots}}}}, \quad (8.8)$$

where b_m is given by (8.7). Equivalently, with $a_m = -b_m$,

$$W_0(x) = \frac{x}{1 + \mathbf{K}_{m=2}^{\infty} \frac{-a_m x}{1 + a_m x}}. \quad (8.9)$$

Every finite convergent in the Euler construction equals the corresponding partial sum of the Taylor series.

Proof. Equation (8.6) says that the ratio entering Euler's fraction at level $m-1$ is $-b_m x$. Substitution into (8.4) gives the finite version of (8.8). By theorem 8.3, its value is the Taylor partial sum. Those partial sums converge to $W_0(x)$ for $|x| < 1/e$ by theorem 4.2; therefore the continued-fraction convergents have the same limit. $\square$

The first coefficients are $b_2 = 1$, $b_3 = 3/2$, $b_4 = 16/9$, and $b_5 = 125/64$. This particular fraction is an exact reorganization of the Taylor series, not a claim that every continued fraction converges beyond the series disk. More sophisticated C - and J -fractions can have different convergence behavior; see [4].

8.4 Schröder-type inverse series

Newton's method uses only the first term of a local series for the correction to an approximate root. The whole series can be generated by Lagrange inversion.

Theorem 8.5 (General inverse correction series). *Let F be analytic near a , with $F'(a) \neq 0$, and suppose a simple root $r = a + h$ lies sufficiently close to a . Set*

$$q = \frac{F(a)}{F'(a)}, \quad A(t) = \frac{F(a+t) - F(a)}{F'(a)t}, \quad (8.10)$$

where $A(0) = 1$. Then

$$h = \sum_{n=1}^{\infty} \frac{(-q)^n}{n} [t^{n-1}] A(t)^{-n}. \quad (8.11)$$

In particular, if

$$a_2 = \frac{F''(a)}{2F'(a)}, \quad a_3 = \frac{F'''(a)}{6F'(a)},$$

then

$$h = -q - a_2 q^2 + (a_3 - 2a_2^2) q^3 + \mathcal{O}(q^4). \quad (8.12)$$

Proof. The root equation $F(a+h) = 0$ and (8.10) give

$$0 = F(a) + F'(a)hA(h), \quad h = \frac{-q}{A(h)}.$$

This has Lagrange form $h = z\phi(h)$ with $z = -q$ and $\phi(t) = A(t)^{-1}$. Formula (8.11) follows from theorem 4.1 with $F(t) = t$.

Now $A(t) = 1 + a_2 t + a_3 t^2 + \mathcal{O}(t^3)$. Expanding $A(t)^{-n}$ in (8.11) gives, successively, $-q$, $-a_2 q^2$, and $(a_3 - 2a_2^2) q^3$, proving (8.12). $\square$

For the Lambert equation $F(w) = we^w - x$, the normalized function is explicit:

$$A_a(t) = \frac{(a+t)e^t - a}{(1+a)t}. \quad (8.13)$$

Also

$$a_2 = \frac{a+2}{2(a+1)}, \quad a_3 = \frac{a+3}{6(a+1)}. \quad (8.14)$$

The first term $h = -q$ is Newton's correction. Keeping more terms produces Schröder-type high-order approximations. Howard's work in the supplied archive develops such series to high order for both W_0 and W_{-1} and studies branch-compatible starting functions [11].

8.5 The Wright omega function

Exponentiating a huge number merely to call W can be undesirable. Define the Wright omega function by

$$\omega(z) = W(e^z). \quad (8.15)$$

On the real axis the unambiguous choice is $\omega(y) = W_0(e^y) > 0$. It satisfies

$$\omega(y) + \log \omega(y) = y. \quad (8.16)$$

Proposition 8.6. *For real y , the equation (8.16) has exactly one positive solution. It is strictly increasing and*

$$\omega'(y) = \frac{\omega(y)}{1 + \omega(y)}. \quad (8.17)$$

Proof. The function $g(w) = w + \log w$ is strictly increasing from $-\infty$ to $+\infty$ on $(0, \infty)$ because $g'(w) = 1 + 1/w > 0$. It therefore has a unique positive inverse. Differentiating (8.16) gives $\omega'(1 + 1/\omega) = 1$, which is (8.17); it is positive. $\square$

The omega formulation is the logarithmic equation used later for stable numerical evaluation. It also removes one layer of exponential growth from asymptotic analysis.

9 Numerical evaluation

A reliable evaluator must solve three different problems:

1. choose the correct branch;
2. obtain a starting value in a safe region;
3. refine without overflow, loss of branch, or amplification of the intrinsic branch-point conditioning.

No single formula is best everywhere. Near $-1/e$ the signed- p series is natural; near 0^- on W_{-1} the logarithmic asymptotic series is natural; away from singular points, high-order iteration is efficient.

9.1 The simple logarithmic fixed-point iteration

From (1.5), any nonzero real branch value satisfies

$$W = \log \left(\frac{x}{W} \right), \quad (9.1)$$

where $x/W > 0$. This suggests

$$w_{n+1} = G_x(w_n), \quad G_x(w) = \log \left(\frac{x}{w} \right). \quad (9.2)$$

It is attractive because it is simple, but its convergence is branch- and range-dependent.

Proposition 9.1 (Local behavior of logarithmic fixed-point iteration). *At a real fixed point $W \neq 0$,*

$$G'_x(W) = -\frac{1}{W}. \quad (9.3)$$

Thus the iteration is locally attracting when $|W| > 1$ and locally repelling when $|W| < 1$. In particular, it is locally attracting on W_0 only for $x > e$, and on W_{-1} for every $-1/e < x < 0$; it loses contraction at the branch point and at $W_0(e) = 1$.

Proof. Differentiate $G_x(w) = \log|x| - \log|w|$ to get $G'_x(w) = -1/w$. The elementary fixed-point theorem says that a differentiable map is locally contractive when the derivative magnitude at the fixed point is less than 1, and repelling when it is greater than 1. Translate $|W_0(x)| > 1$ and $|W_{-1}(x)| > 1$ using the branch ranges and special values. $\square$

Even where it converges, (9.2) is only linear. The next methods are quadratic or cubic.

9.2 Newton, Halley, and logarithmic Newton

For

$$f_x(w) = we^w - x,$$

Newton's method is

$$N_x(w) = w - \frac{we^w - x}{e^w(1+w)} = w - \frac{w - xe^{-w}}{1+w}. \quad (9.4)$$

Halley's method uses the second derivative:

$$H_x(w) = w - \frac{f_x(w)}{f'_x(w) - \frac{f_x(w)f''_x(w)}{2f'_x(w)}}, \quad f'_x(w) = e^w(1+w), \quad f''_x(w) = e^w(2+w). \quad (9.5)$$

A particularly useful alternative applies Newton's method to the logarithmic equation

$$F_x(w) = w + \log|w| - \log|x| = 0. \quad (9.6)$$

Its update is

$$\begin{aligned} S_x(w) &= w - \frac{w + \log|w| - \log|x|}{1 + 1/w} \\ &= \frac{w}{1+w} \left(1 + \log \frac{x}{w} \right). \end{aligned} \quad (9.7)$$

The quotient x/w is positive on either real branch. Formula (9.7) avoids constructing e^w , an important advantage when x and $W_0(x)$ are enormous.

Theorem 9.2 (Local error equations). *Let $W \neq -1, 0$ be a simple real branch value and let $e = w - W$. One step of the three methods has the following local errors:*

$$N_x(w) - W = \frac{W+2}{2(W+1)}e^2 + \mathcal{O}(e^3), \quad (9.8)$$

$$S_x(w) - W = -\frac{1}{2W(W+1)}e^2 + \mathcal{O}(e^3), \quad (9.9)$$

$$H_x(w) - W = \frac{W^2 + 4W + 6}{12(W+1)^2}e^3 + \mathcal{O}(e^4). \quad (9.10)$$

Thus Newton and logarithmic Newton are quadratic and Halley is cubic, except where the displayed denominators reveal the branch-point singularity.

Proof. For a smooth function q with simple root W , Taylor expansion of Newton's map gives the standard formula

$$e_{\text{new}} = \frac{q''(W)}{2q'(W)}e^2 + \mathcal{O}(e^3). \quad (9.11)$$

To verify it directly, write $q(W + e) = q'(W)e + q''(W)e^2/2 + \mathcal{O}(e^3)$ and $q'(W + e) = q'(W) + q''(W)e + \mathcal{O}(e^2)$, divide the two series, and subtract from e .

For $q = f_x$, the ratio is $f_x''(W)/f_x'(W) = (W + 2)/(W + 1)$, proving (9.8). For $q = F_x$,

$$F_x'(W) = \frac{W + 1}{W}, \quad F_x''(W) = -\frac{1}{W^2},$$

which proves (9.9).

For Halley's method, a direct division of Taylor series gives

$$e_{\text{new}} = \left(\frac{q''(W)^2}{4q'(W)^2} - \frac{q'''(W)}{6q'(W)} \right) e^3 + \mathcal{O}(e^4). \quad (9.12)$$

For f_x , one has $f_x'''(W) = e^W(W + 3)$. Substitution into (9.12) gives

$$\frac{(W + 2)^2}{4(W + 1)^2} - \frac{W + 3}{6(W + 1)} = \frac{W^2 + 4W + 6}{12(W + 1)^2},$$

proving (9.10). □

Remark 9.3 (Which Newton form is faster?). For large positive W , the coefficient in (9.8) tends to $1/2$, while the coefficient in (9.9) is of order $1/W^2$. Logarithmic Newton therefore becomes dramatically more accurate per step at large x . Near $W = 0$, however, its coefficient is large and ordinary Newton is preferable. This quantitative comparison was emphasized by Iacono and Boyd [12].

9.3 Monotone quadratic evaluation of W_0 for $x > e$

The two-term lower bound in theorem 6.5 gives a starting value that never crosses the root.

Theorem 9.4 (Monotone logarithmic Newton on W_0). *Let $x > e$, set*

$$\beta_0 = \log x - \log \log x, \quad (9.13)$$

and define $\beta_{n+1} = S_x(\beta_n)$ by (9.7). Then

$$1 < \beta_0 < \beta_1 < \cdots < W_0(x), \quad (9.14)$$

and $\beta_n \rightarrow W_0(x)$. If $w \in (1, W_0(x))$, then

$$0 < W_0(x) - S_x(w) < \frac{(W_0(x) - w)^2}{(1 + w)W_0(x)}. \quad (9.15)$$

Proof. Let $W = W_0(x) > 1$. By theorem 6.5, $\beta_0 < W$. Also, for $L = \log x > 1$, the function $L - \log L$ has minimum 1 at $L = 1$, so $\beta_0 > 1$.

Fix $1 < w < W$ and put

$$r = \frac{W - w}{w} > 0.$$

Using $\log x = W + \log W$ in (9.7), a direct simplification gives

$$W - S_x(w) = \frac{w}{1 + w}(r - \log(1 + r)). \quad (9.16)$$

For $r > 0$,

$$0 < r - \log(1 + r) < \frac{r^2}{1 + r}.$$

The first inequality follows from $\log(1 + r) < r$. For the second, subtract the left side from the proposed upper bound:

$$\frac{r^2}{1 + r} - r + \log(1 + r) = \log(1 + r) - \frac{r}{1 + r}.$$

This expression is zero at 0 and has derivative $r/(1 + r)^2 > 0$. Substitution into (9.16) proves (9.15); in particular $S_x(w) < W$.

Moreover,

$$S_x(w) - w = \frac{w}{1 + w}(W - w + \log(W/w)) > 0,$$

so every iterate increases and remains below W . It therefore converges to some limit $\ell \leq W$. Continuity gives $S_x(\ell) = \ell$, which is $\ell + \log \ell = \log x$. This equation has a unique positive solution, so $\ell = W$. $\square$

9.4 Monotone quadratic evaluation of W_{-1}

The lower branch also admits a global monotone scheme, but the direction is easiest to see through concavity of the logarithmic equation.

Theorem 9.5 (Monotone logarithmic Newton on W_{-1}). *Let $-1/e < x < 0$, put $u = -1 - \log(-x)$, and start with*

$$\beta_0 = -1 - \sqrt{2u} - u.$$

Define $\beta_{n+1} = S_x(\beta_n)$. Then

$$\beta_0 < \beta_1 < \cdots < W_{-1}(x) < -1, \tag{9.17}$$

and $\beta_n \rightarrow W_{-1}(x)$ quadratically.

Proof. Let

$$F(w) = w + \log(-w) - \log(-x), \quad w < -1.$$

Then

$$F'(w) = 1 + \frac{1}{w} = \frac{w + 1}{w} > 0, \quad F''(w) = -\frac{1}{w^2} < 0.$$

Thus F is strictly increasing and strictly concave, with root $W = W_{-1}(x)$. By theorem 7.3, $\beta_0 < W$.

Suppose $w < W$. Since F is increasing, $F(w) < 0$, so the Newton update $S_x(w) = w - F(w)/F'(w)$ is greater than w . Concavity says that the graph lies below its tangent:

$$F(W) \leq F(w) + F'(w)(W - w).$$

Since $F(W) = 0$, this implies $-F(w)/F'(w) \leq W - w$, hence $S_x(w) \leq W$. Strict concavity gives strict inequality. Induction proves (9.17). The increasing bounded sequence has a limit; continuity and uniqueness of the root force the limit to be W . Quadratic convergence follows from (9.9), because $W \neq -1, 0$ in the open domain. $\square$

9.5 Guaranteed branch-safe refinement

Monotone iterations are attractive, but a general library routine benefits from a bracket and a safeguarded high-order step.

Algorithm 9.6 (Safeguarded Halley–bisection). For a requested real branch and argument:

1. Handle $x = -1/e$ and $x = 0$ exactly.
2. Construct a bracket $[a, b]$:
 - for W_0 with $-1/e < x < 0$, use $[-1, 0]$;
 - for W_0 with $x > 0$, use $[x/(1+x), \log(1+x)]$;
 - for W_{-1} , use the two bounds in (7.7).
3. From a current point w , compute a Halley candidate $h = H_x(w)$. Accept it only if it is finite and lies in the middle half $[a + (b-a)/4, b - (b-a)/4]$; otherwise use the midpoint $h = (a+b)/2$.
4. Update the bracket using monotonicity of $f(w) = we^w$: on W_0 , replace a by h if $f(h) < x$ and replace b otherwise; on W_{-1} , reverse this test because f is decreasing.
5. Stop when the bracket width or the residual is below the requested tolerance.

Theorem 9.7 (Convergence of the safeguarded algorithm). *Algorithm 9.6 preserves a bracket containing the requested branch value and converges to it. Every accepted Halley step taken sufficiently close to a non-branch-point root obeys the cubic local error law in (9.10); the unconditional global guarantee is geometric contraction of the bracket.*

Proof. The initial brackets are valid by (7.3), theorem 7.2, and theorem 7.3. The update rule is exactly the comparison principle theorem 7.6, so the root remains inside every new bracket.

An accepted candidate lies in the middle half, so whichever endpoint is replaced, the new width is at most $3/4$ of the old width. A rejected candidate is replaced by the midpoint, which halves the width. Therefore the width tends to zero. The nested closed brackets have one common point, and continuity plus the residual sign change identifies it as the branch root.

At a simple root, theorem 9.2 proves that any Halley step accepted in the local regime has cubic error. The middle-half safeguard may still insert occasional bisections when the root lies very near one endpoint of the current bracket, so cubic convergence is asserted for accepted local Halley steps rather than as an unconditional rate for the hybrid sequence. The bracket contraction already proves global convergence. $\square$

9.6 Analytic starting approximations

For $x > 0$, Iacono and Boyd propose

$$\alpha(x) = \frac{1}{1 + \frac{1}{2} \log(1+x)}, \quad G_+(x) = \log(1 + \alpha(x)x). \quad (9.18)$$

It reproduces the first two Taylor terms at 0 and the first two logarithmic terms at infinity. It is an approximation, not a rigorous enclosure.

A single expression covering the entire principal branch is built from $y = \sqrt{1+ex}$:

$$G_0(x) = -1 + a \log \left(\frac{1+by}{1+c \log(1+y)} \right), \quad (9.19)$$

where

$$a = 2.036, \quad c = \frac{e^{1/a} - 1 - \sqrt{2}/a}{1 - e^{1/a} \log 2}, \quad b = \frac{\sqrt{2}}{a} + c. \quad (9.20)$$

The printed decimal values are approximately $b = 1.14956131$ and $c = 0.45495740$. Writing them through (9.20) makes $G_0(0) = 0$ exact instead of introducing a tiny rounding offset. The square-root variable gives the correct singular shape at $-1/e$, while the logarithm gives the correct growth at infinity.

For W_{-1} , three branch-aware choices cover the domain:

- use the negative signed- p series near $-1/e$;
- use the lower rigorous bound (7.11) when a guaranteed monotone iteration is desired;
- use the logarithmic asymptotic expansion (6.10) near 0^- .

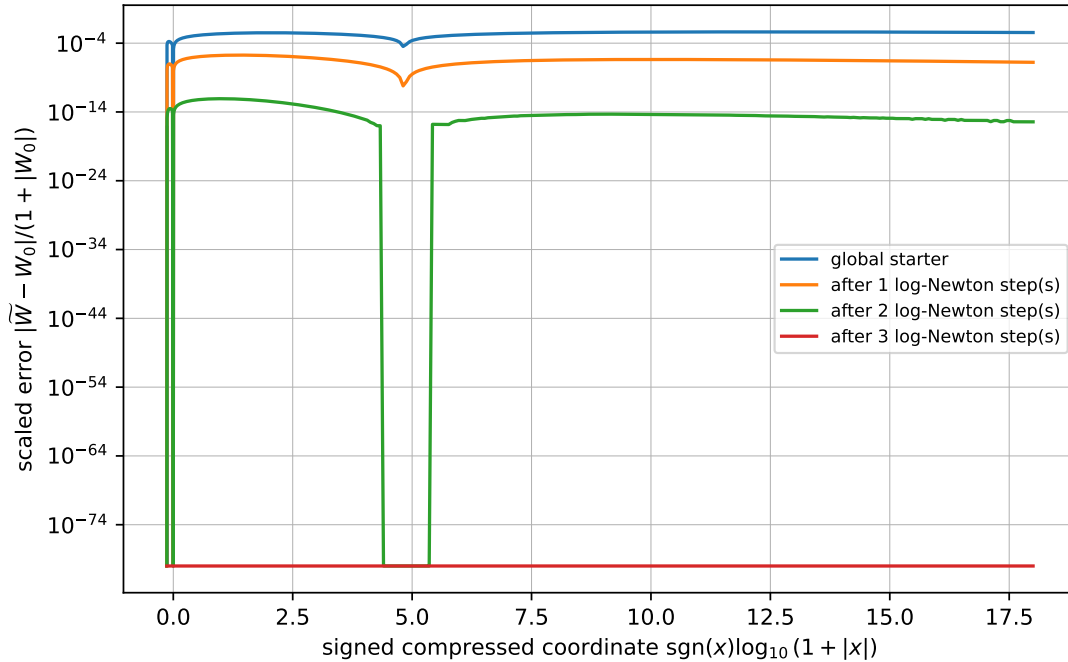


Figure 6: Sampled scaled error of the global principal-branch starter and one, two, and three logarithmic Newton corrections. The coordinate compresses the whole interval $[-1/e, \infty)$.

9.7 Reproducible numerical experiments

The accompanying script `lambert_w_experiments.py` uses 80-digit `mpmath` reference values. It generates all numerical figures, the CSV table in `data/numerical_results.csv`, and the following LaTeX table. The maxima are sampled values on explicitly stated grids, not computer-assisted proofs of uniform error bounds.

The distinction matters. A dense grid can falsify an overoptimistic claim and measure practical behavior, but only analytic inequalities or interval arithmetic can certify a continuum. Lóczi’s work develops guaranteed high-precision enclosures and stopping rules for both real branches [15]; the safeguarded construction above gives a simpler proof of the same guiding principle: never sacrifice a valid branch bracket merely for a fast local step.

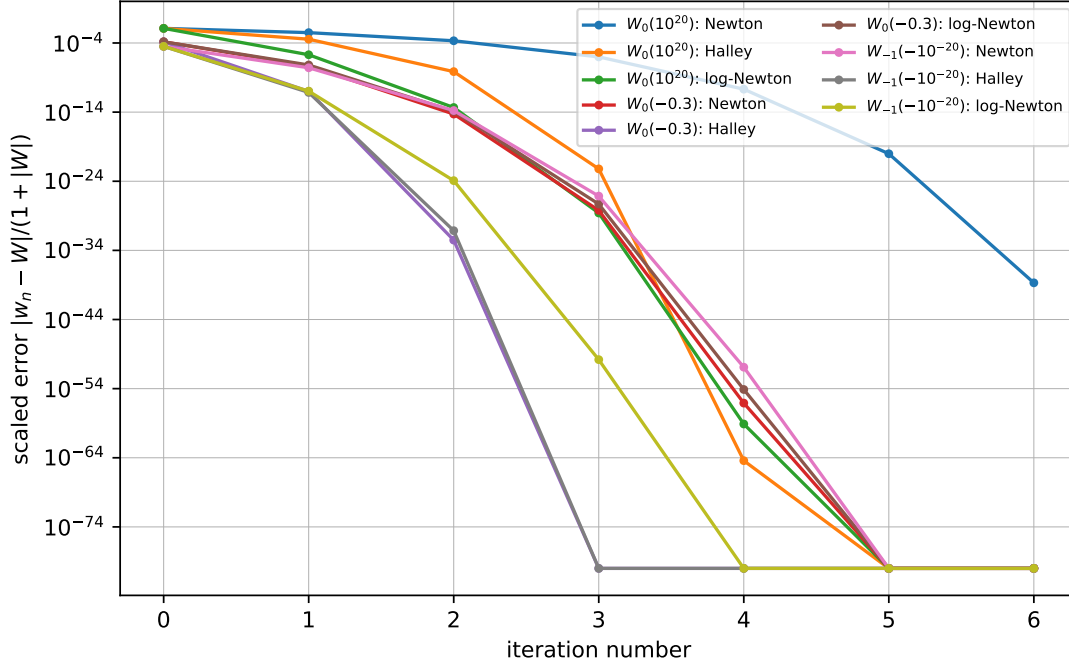


Figure 7: Representative high-precision experiments comparing ordinary Newton, Halley, and logarithmic Newton. Local orders are exactly those proved in theorem 9.2; the constants determine which method wins in a particular region.

Table 2: Sampled numerical errors. These are reproducible experimental maxima on the stated grids, not certified uniform bounds.

Approximation or quantity	Sampled domain	Measure	Maximum
Iacono–Boyd positive starter	$10^{-12} \leq x \leq 10^{18}$	relative	3.532×10^{-2}
positive starter + 1 log-Newton step(s)	$10^{-12} \leq x \leq 10^{18}$	relative	2.161×10^{-4}
positive starter + 2 log-Newton step(s)	$10^{-12} \leq x \leq 10^{18}$	relative	1.225×10^{-8}
positive starter + 3 log-Newton step(s)	$10^{-12} \leq x \leq 10^{18}$	relative	4.012×10^{-17}
Iacono–Boyd global starter	$-1/e + 10^{-14}/e \leq x \leq 10^{18}$	scaled	4.353×10^{-3}
[2/2] Padé approximant	$-0.25 \leq x \leq 0.25$	scaled	1.361×10^{-3}
Puiseux series through p^8 on W_0	$10^{-14} \leq 1 + ex \leq 0.25$	scaled	1.294×10^{-4}
Puiseux series through p^8 on W_{-1}	$10^{-14} \leq 1 + ex \leq 0.25$	scaled	1.637×10^{-4}
four-correction asymptotic expansion of W_0	$10^6 \leq x \leq 10^{50}$	relative	8.856×10^{-8}
four-correction asymptotic expansion of W_{-1}	$10^{-60} \leq x \leq 10^{-6}$	relative	1.116×10^{-7}
relative width of the rigorous W_{-1} bracket	$10^{-10} \leq u = -1 - \log(-x) \leq 10^{10}$	scaled width	3.333×10^{-1}

10 Complex branches

The real picture is the visible edge of a multivalued complex inverse. This section gives a working introduction without requiring a full course in Riemann surfaces.

10.1 Local inverses and the exceptional value

Let

$$f(w) = we^w, \quad w \in \mathbb{C}.$$

Its complex derivative is still

$$f'(w) = e^w(1 + w).$$

Because the exponential never vanishes, the only critical point is $w = -1$, and its critical value is $-1/e$.

Theorem 10.1 (Local complex inverse). *If w_* satisfies $w_*e^{w_*} = z_*$ and $w_* \neq -1$, then there are neighborhoods U of w_* and V of z_* such that $f : U \rightarrow V$ is one-to-one and has an analytic inverse. The derivative of that inverse is*

$$\frac{e^{-w}}{1 + w}.$$

At $w_ = -1$ no analytic inverse in the variable $z + 1/e$ exists; the local inverse instead has the square-root form of theorem 5.1.*

Proof. The complex inverse-function theorem says that an analytic function with nonzero derivative at a point is one-to-one in a sufficiently small neighborhood and has an analytic local inverse. Here $f'(w_*) = e^{w_*}(1 + w_*) \neq 0$ exactly when $w_* \neq -1$. Differentiating the identity $f(W(z)) = z$ gives $W'(z) = 1/f'(W(z)) = e^{-W(z)}/(1 + W(z))$.

At $w_* = -1$, put $u = w + 1$ and $\zeta = z + 1/e$. Equation (5.3) can be written

$$e\zeta = \frac{u^2}{2}h(u),$$

where h is analytic and $h(0) = 1$. Suppose an ordinary analytic inverse $u = u(\zeta)$ existed with $u(0) = 0$. If its zero at 0 had order $m \geq 1$, then $u(\zeta)^2h(u(\zeta))$ would have a zero of even order $2m$, whereas $e\zeta$ has a simple zero. This contradiction rules out an ordinary analytic inverse. Taking a square root instead produces the two local inverses constructed explicitly in theorem 5.1. $\square$

10.2 Branch labels and cuts

The standard branches are denoted W_k , $k \in \mathbb{Z}$, but branch cuts are conventions rather than intrinsic objects. We follow the convention of Corless et al. [5]. The principal branch W_0 is normalized by $W_0(0) = 0$ and is analytic on the plane cut along $(-\infty, -1/e]$. The branches W_{-1} and W_1 are exceptional: each has a logarithmic branch point at 0 and also shares the square-root branch point at $-1/e$. They are represented with two leftward cuts, one issuing from each branch point; the cuts overlap on $(-\infty, -1/e)$. The remaining nonprincipal branches have the logarithmic cut $(-\infty, 0]$. Boundary values on any cut must be specified by approaching from above or below. With the standard closure convention, W_{-1} takes the real values described earlier on $[-1/e, 0)$.

Why are integers natural labels? Taking a logarithm of $we^w = z$ gives, with branch choices,

$$w + \log w = \text{Log } z + 2\pi i k. \quad (10.1)$$

For large $|k|$ or large $|z|$, the dominant first approximation is the right-hand side, so one solution lies near each vertical translate by $2\pi i k$.

10.3 Asymptotics on every complex branch

Let Log denote the principal complex logarithm. In a standard cut-free sector, the logarithmic equation for branch k is organized by

$$L_1 = \text{Log } z + 2\pi ik, \quad L_2 = \text{Log } L_1. \quad (10.2)$$

Along a more general continuation path, the logarithms are chosen continuously and the integer on the right of (10.1) may require a constant relabeling. What matters below is that L_1 is the continuous right-hand side of that equation. The same coefficient polynomials from the real theory reappear.

Theorem 10.2 (Complex branch asymptotics). *Follow a branch $W_k(z)$ along a cut-free path on which $|W_k(z)| \rightarrow \infty$. Choose continuous logarithms so that*

$$W_k(z) + \log W_k(z) = L_1, \quad L_2 = \log L_1. \quad (10.3)$$

Then $|L_1| \rightarrow \infty$, $L_2/L_1 \rightarrow 0$, and, for every fixed N ,

$$W_k(z) = L_1 - L_2 + \sum_{j=1}^N \frac{P_j(L_2)}{L_1^j} + \mathcal{O}\left(\frac{|L_2|^{N+1}}{|L_1|^{N+1}}\right), \quad (10.4)$$

with P_j given by (6.5). In particular,

$$W_k(z) = L_1 - L_2 + \frac{L_2}{L_1} + \cdots.$$

In the usual large- z sectors, L_1 and L_2 may be taken as in (10.2).

Proof. Write $W = W_k(z)$. From (10.3), $L_1 = W + \log W$. Since $|W| \rightarrow \infty$, $\log W/W \rightarrow 0$ along the chosen determination, so $L_1/W \rightarrow 1$. Hence $|L_1| \rightarrow \infty$ and $L_2/L_1 \rightarrow 0$.

Substitute $W = L_1 - L_2 + v$, $t = 1/L_1$, and $L = L_2$. Since

$$W = L_1(1 - tL + tv),$$

the selected logarithm satisfies, for all sufficiently advanced points on the path,

$$\log W = L_2 + \log(1 - tL + tv).$$

Equation (10.3) therefore becomes exactly (6.12). The Lagrange-inversion calculation in the proof of theorem 6.1 is algebraic over the complex numbers and gives the same polynomials P_j .

It remains to justify the remainder without using a nonexistent complex mean value theorem. Let

$$w_N = L_1 - L_2 + \sum_{j=1}^N \frac{P_j(L_2)}{L_1^j}$$

and let

$$R_N = w_N + \log w_N - L_1.$$

The same coefficient cancellation as in the real proof gives

$$R_N = \mathcal{O}\left(\frac{|L_2|^{N+1}}{|L_1|^{N+1}}\right).$$

Seek the exact root as $W = w_N + \delta$. On the same local logarithm sheet, its equation is

$$\delta = -R_N - \log \left(1 + \frac{\delta}{w_N} \right). \quad (10.5)$$

Because $w_N \sim L_1$ and $R_N \rightarrow 0$, for sufficiently large $|L_1|$ the right-hand side of (10.5) maps the closed disk $|\delta| \leq 2|R_N|$ into itself. Indeed, when $|\zeta| \leq 1/2$, the logarithm series gives $|\log(1 + \zeta)| \leq 2|\zeta|$, and therefore

$$\left| -R_N - \log \left(1 + \frac{\delta}{w_N} \right) \right| \leq |R_N| + \frac{2|\delta|}{|w_N|} \leq 2|R_N|$$

for all sufficiently large $|w_N|$. Its derivative with respect to δ has magnitude $1/|w_N + \delta| < 1/2$ there. The contraction principle supplies a unique fixed point with $|\delta| \leq 2|R_N|$. It is the continuation of the chosen branch, and the required error estimate follows. $\square$

For fixed nonzero z , letting $|k| \rightarrow \infty$ describes the remote branches. For fixed k , the expansion applies as $|z| \rightarrow \infty$ in sectors avoiding the cuts. As $z \rightarrow 0$, it applies to the nonprincipal branches whose magnitudes diverge; it does *not* apply to $W_0(z)$, which instead satisfies $W_0(z) = z + \mathcal{O}(z^2)$.

10.4 Conjugation symmetry

Proposition 10.3 (Conjugate branches). *Away from the conventional branch cuts, standard branch labels satisfy*

$$\overline{W_k(z)} = W_{-k}(\bar{z}). \quad (10.6)$$

In particular, $W_0(\bar{z}) = \overline{W_0(z)}$ wherever both sides are continued without crossing the principal cut.

Proof. If $w^w = z$, then complex conjugation gives $\overline{w^w} = \bar{z}$. Hence conjugation maps inverse values at z to inverse values at $\bar{z}$. To identify the label, use (10.4): conjugation changes $\text{Log } z + 2\pi i k$ into $\text{Log } \bar{z} - 2\pi i k$ along conjugate paths, which is the leading quantity for branch $-k$. Analytic continuation preserves this identification through any cut-free connected region. $\square$

Warning 10.4. On a branch cut, an expression such as $W_k(x)$ may mean an upper or lower boundary value. Formula (10.6) then relates opposite boundary values, not necessarily two symbols interpreted with the same one-sided convention. The real W_{-1} on $(-1/e, 0)$ is best treated directly by the real inverse construction of theorem 1.2.

10.5 The complex square-root interchange

Near $z = -1/e$, let

$$p = \sqrt{2(1 + ez)}$$

with either local square-root choice. Formula (5.7) gives two analytic functions of p . Circling once around $-1/e$ changes p to $-p$ and interchanges the two local inverse sheets. On the real axis to the right of the branch point, those sheets are precisely W_0 and W_{-1} . This is the geometric meaning of a square-root branch point.

11 Applications

The point of a special function is not to rename an equation; it is to expose structure. A Lambert W solution makes branch counts, derivatives, asymptotics, and parameter dependence available without resolving the transcendental equation from scratch.

11.1 The equation $x^x = a$

Theorem 11.1 (Complete positive-real classification). *Let $a > 0$ and seek solutions $x > 0$ of*

$$x^x = a. \quad (11.1)$$

They are

$$x = \exp(W_k(\log a)), \quad (11.2)$$

with the following real count:

Range of a	Positive solutions
$0 < a < e^{-1/e}$	none
$a = e^{-1/e}$	one: $x = 1/e$
$e^{-1/e} < a < 1$	two: branches $k = 0, -1$
$a = 1$	one: $x = 1$
$a > 1$	one: branch $k = 0$

Proof. Put $y = \log x$, so $x = e^y$. Taking a logarithm of (11.1) gives

$$x \log x = \log a, \quad ye^y = \log a.$$

Thus $y = W_k(\log a)$ and (11.2) follows. The real count is exactly the branch count for the argument $\log a$: there are no real values below $-1/e$, one at $-1/e$, two between $-1/e$ and 0, and one above 0. At $a = 1$, the finite value is $W_0(0) = 0$, giving $x = 1$; $W_{-1}(0)$ is not finite and does not supply an additional positive solution. $\square$

This theorem recovers the familiar minimum of x^x on $(0, \infty)$ without a separate optimization: the two real branches merge exactly when $\log a = -1/e$.

11.2 Infinite power towers

For $a > 0$, define the finite towers by

$$t_0 = 1, \quad t_{n+1} = a^{t_n}. \quad (11.3)$$

If t_n converges to T , then $T = a^T$.

Theorem 11.2 (Convergence interval for real power towers). *The sequence (11.3) converges to a finite positive limit if and only if*

$$e^{-e} \leq a \leq e^{1/e}. \quad (11.4)$$

On this interval its limit is

$$T = \begin{cases} \frac{W_0(-\log a)}{-\log a}, & a \neq 1, \\ 1, & a = 1. \end{cases} \quad (11.5)$$

For $1 < a < e^{1/e}$ there is also a larger, unstable fixed point obtained from $W_{-1}(-\log a)$, but the tower beginning at 1 converges to the W_0 fixed point.

Proof. Write $b = \log a$ and $g(x) = e^{bx}$. A positive fixed point T satisfies $\log T = bT$, or

$$(-bT)e^{-bT} = -b. \quad (11.6)$$

This gives (11.5). We now prove the exact convergence interval.

Case 1: $b \geq 0$. A fixed point has $T \geq 1$ and satisfies

$$b = \frac{\log T}{T}.$$

The function $\phi(T) = \log T/T$ increases from 0 to $1/e$ on $[1, e]$ and then decreases to 0, because $\phi'(T) = (1 - \log T)/T^2$. Hence a fixed point exists only if $b \leq 1/e$. When $0 \leq b \leq 1/e$, let $T \in [1, e]$ be the smaller fixed point. Since g is increasing, $t_0 = 1 \leq T$ implies $t_n \leq T$ for all n . Also $t_1 \geq t_0$, and monotonicity of g then gives $t_{n+1} \geq t_n$. Thus t_n increases and is bounded, so it converges; its limit must be the fixed point T . If $b > 1/e$, no fixed point exists, so a finite limit is impossible.

Case 2: $b < 0$. Put $c = -b > 0$, so $g(x) = e^{-cx}$ is decreasing. There is a unique fixed point $T \in (0, 1)$ because

$$c = \frac{-\log T}{T}$$

and the right side decreases continuously from $+\infty$ to 0 on $(0, 1)$. The condition $c \leq e$ is equivalent to $T \geq 1/e$.

Assume first $c \leq e$. Since $t_0 = 1 > T$, decreasingness gives $t_1 = g(t_0) < g(T) = T$. It follows inductively that the even subsequence lies above T and the odd subsequence below T . Moreover, $t_2 < t_0$ and $t_3 > t_1$; because g reverses inequalities, the even subsequence decreases and the odd subsequence increases. They therefore have limits $q \geq T$ and $p \leq T$, satisfying

$$p = e^{-cq}, \quad q = e^{-cp}. \quad (11.7)$$

We show that $p = q$ whenever $c \leq e$. Suppose instead $0 < p < q$ and write $r = q/p > 1$, $s = \log r > 0$. Taking logarithms in (11.7) gives

$$s = c(q - p) = cp(r - 1), \quad \log p = -cpr = -\frac{rs}{r - 1}.$$

Thus

$$c = \frac{s}{e^s - 1} \exp\left(\frac{se^s}{e^s - 1}\right). \quad (11.8)$$

Let $H(s)$ be the logarithm of the right side divided by e :

$$H(s) = \log \frac{s}{e^s - 1} + \frac{se^s}{e^s - 1} - 1.$$

A calculation gives

$$H'(s) = \frac{(e^s - 1)^2 - s^2 e^s}{s(e^s - 1)^2} > 0,$$

because $e^s - 1 = 2e^{s/2} \sinh(s/2) > se^{s/2}$ for $s > 0$. Also $H(s) \rightarrow 0$ as $s \rightarrow 0^+$. Hence $H(s) > 0$, so (11.8) implies $c > e$, a contradiction. Therefore $p = q = T$ and the entire sequence converges.

Finally suppose $c > e$. Then the fixed point has $T < 1/e$, so

$$|g'(T)| = cT = -\log T > 1.$$

If t_n converged to T , the mean value theorem would give

$$\frac{|t_{n+1} - T|}{|t_n - T|} = |g'(\xi_n)| \longrightarrow |g'(T)| > 1$$

for points ξ_n between t_n and T . Since $t_0 \neq T$, this eventually forces the errors to grow rather than tend to zero, a contradiction. Thus convergence occurs exactly for $c \leq e$, that is, $a = e^{-c} \geq e^{-e}$.

Combining the two cases gives (11.4). The convergent fixed point has $-bT \geq -1$, so it is represented by W_0 in (11.5); for $b > 0$ the second value from W_{-1} corresponds to the larger unstable fixed point. $\square$

11.3 A Wien-type transcendental equation

Many spectral maxima reduce to an equation of the form

$$m(1 - e^{-x}) = x. \quad (11.9)$$

Proposition 11.3 (Branch roles in the Wien equation). *For $m > 0$, every solution of (11.9) is*

$$x = m + W_k(-me^{-m}). \quad (11.10)$$

If $m > 1$, W_{-1} gives the trivial root $x = 0$, while W_0 gives the unique positive root. If $0 < m < 1$, W_0 gives $x = 0$ and the other real branch gives a negative root.

Proof. Equation (11.9) is equivalent to $m - x = me^{-x}$. Put $y = x - m$. Then

$$-y = me^{-m-y}, \quad ye^y = -me^{-m},$$

which proves (11.10). Since $(-m)e^{-m} = -me^{-m}$, the branch whose range contains $-m$ gives $W_k(-me^{-m}) = -m$ and hence $x = 0$. For $m > 1$, $-m < -1$, so this is W_{-1} ; the principal value lies in $(-1, 0)$ and yields $x > m - 1 > 0$. For $0 < m < 1$, $-m \in (-1, 0)$ belongs to the principal range, so W_0 gives the zero root; the lower value is less than -1 and yields a negative root. Uniqueness in each range follows from the two-branch theorem. $\square$

This example is a model branch-selection lesson: blindly using W_0 can return either the physically interesting root or a trivial root, depending on the parameter.

11.4 Delay differential equations

Consider the linear delay equation

$$y'(t) = a y(t - \tau), \quad \tau > 0. \quad (11.11)$$

Seek exponential modes $y(t) = e^{\lambda t}$. Substitution gives

$$\lambda e^{\lambda t} = a e^{\lambda(t-\tau)},$$

so

$$\lambda e^{\lambda \tau} = a, \quad \lambda_k = \frac{1}{\tau} W_k(a\tau). \quad (11.12)$$

Every complex branch contributes a characteristic exponent. Stability is controlled by the real parts of this infinite family, illustrating why the complex branches are not optional in delay problems.

Proposition 11.4. *Every exponential solution of (11.11) is of the form $C \exp(\lambda_k t)$ with (11.12); conversely every such mode solves the equation.*

Proof. The preceding substitution proves necessity. Conversely, if $\lambda_k \tau \exp(\lambda_k \tau) = a\tau$, then $\lambda_k = a \exp(-\lambda_k \tau)$. Therefore

$$\frac{d}{dt} e^{\lambda_k t} = \lambda_k e^{\lambda_k t} = a e^{\lambda_k(t-\tau)},$$

which is (11.11). □

11.5 Michaelis–Menten depletion kinetics

Let a substrate concentration $S(t) > 0$ satisfy

$$\frac{dS}{dt} = -\frac{VS}{K+S}, \quad S(0) = S_0 > 0, \quad (11.13)$$

where $V, K > 0$.

Theorem 11.5 (Closed time course). *The solution of (11.13) is*

$$S(t) = KW_0 \left(\frac{S_0}{K} \exp \left(\frac{S_0}{K} - \frac{Vt}{K} \right) \right). \quad (11.14)$$

Proof. Separate variables:

$$\left(1 + \frac{K}{S} \right) dS = -V dt.$$

Integration and the initial condition give

$$S + K \log S = S_0 + K \log S_0 - Vt.$$

Put $y = S/K$ and rearrange:

$$y + \log y = \frac{S_0}{K} + \log \frac{S_0}{K} - \frac{Vt}{K}.$$

Exponentiating gives

$$ye^y = \frac{S_0}{K} \exp \left(\frac{S_0}{K} - \frac{Vt}{K} \right) > 0.$$

The argument is positive, so the only real branch is W_0 . Multiplication by K proves (11.14). Reversing the calculation verifies that it satisfies the differential equation and initial value. □

This explicit solution is one reason Lambert W is useful in biochemical kinetics [10]: it can be differentiated with respect to parameters, integrated, and fitted without an implicit root solve at every time point.

11.6 A diode with series resistance

A standard generalized diode law is

$$I = I_s \left[\exp \left(\frac{V - RI}{nV_T} \right) - 1 \right], \quad (11.15)$$

where $I_s, R, n, V_T > 0$.

Theorem 11.6 (Explicit diode current). *Equation (11.15) has the physical real solution*

$$I = \frac{nV_T}{R} W_0 \left(\frac{RI_s}{nV_T} \exp \left(\frac{V + RI_s}{nV_T} \right) \right) - I_s. \quad (11.16)$$

Proof. Set $J = I + I_s$. Then (11.15) becomes

$$J = I_s \exp \left(\frac{V - R(J - I_s)}{nV_T} \right) = I_s \exp \left(\frac{V + RI_s}{nV_T} \right) \exp \left(-\frac{RJ}{nV_T} \right).$$

Therefore

$$\left(\frac{RJ}{nV_T} \right) \exp \left(\frac{RJ}{nV_T} \right) = \frac{RI_s}{nV_T} \exp \left(\frac{V + RI_s}{nV_T} \right).$$

The right side is positive, so only W_0 is real. Solve for J and subtract I_s to obtain (11.16). $\square$

This closed form, used in transistor and diode circuit analysis [1], smoothly connects the exponential low-current region to the resistance-dominated high-current region.

11.7 Asymptotic inversion in prime-number estimates

The prime number theorem suggests the rough relation

$$n \approx \frac{x}{\log x} \quad (11.17)$$

when x is near the n th prime. Solving this model exactly requires the lower real branch.

Proposition 11.7 (Lambert proxy for inverse logarithmic growth). *For $n > e$, the solution $x > e$ of $n = x / \log x$ is*

$$x = -nW_{-1}(-1/n). \quad (11.18)$$

As $n \rightarrow \infty$,

$$-nW_{-1}(-1/n) = n \left(\log n + \log \log n + \frac{\log \log n}{\log n} + \dots \right). \quad (11.19)$$

Proof. Put $y = \log x > 1$, so $x = e^y$. Equation $n = e^y / y$ becomes

$$(-y)e^{-y} = -\frac{1}{n}.$$

Because $-y < -1$, the correct branch is W_{-1} : $-y = W_{-1}(-1/n)$. Hence

$$x = e^y = e^{-W_{-1}(-1/n)}.$$

Using $We^W = -1/n$ gives $e^{-W} = -nW$, proving (11.18). Formula (11.19) follows by substituting $L_1 = -\log n$ and $L_2 = \log \log n$ into theorem 6.1 and multiplying by $-n$. $\square$

This is a proxy, not an exact formula for primes. It correctly captures the first two growing terms in the scale of the n th prime; refined number-theory approximations supply additional constants and corrections. The branch choice is nevertheless exact and informative: $W_0(-1/n)$ would correspond to a small solution with $\log x < 1$, not the large inverse needed in prime asymptotics [17].

11.8 A general modeling lesson

In each example the decisive step was not “apply W ” but rather:

1. identify a transformed variable u ;
2. reach $ue^u = z$ by reversible algebra;
3. determine the admissible range of u ;
4. select W_0 , W_{-1} , or a complex W_k from that range.

This branch-first viewpoint is especially important in physical and biological models, where one branch can represent a trivial, unstable, negative, or otherwise inadmissible solution.

12 Further analytic consequences

This section collects results that follow systematically from the preceding machinery and are useful when Lambert W appears inside a larger calculation.

12.1 Taylor expansion about an arbitrary regular point

Let $x_0 = w_0 e^{w_0}$ with $w_0 \neq -1$. The branch through (x_0, w_0) is analytic, so theorem 3.2 immediately gives all local coefficients.

Proposition 12.1 (Regular-point Taylor series). *For h sufficiently small,*

$$\begin{aligned} W(x_0 + h) = w_0 + \frac{e^{-w_0}}{1 + w_0} h - \frac{e^{-2w_0}(w_0 + 2)}{2(1 + w_0)^3} h^2 \\ + \frac{e^{-3w_0}(2w_0^2 + 8w_0 + 9)}{6(1 + w_0)^5} h^3 + \mathcal{O}(h^4), \end{aligned} \quad (12.1)$$

and in general

$$[h^n]W(x_0 + h) = \frac{e^{-nw_0} P_n(w_0)}{n!(1 + w_0)^{2n-1}}. \quad (12.2)$$

Proof. Taylor’s theorem gives $W(x_0 + h) = \sum_{n \geq 0} W^{(n)}(x_0) h^n / n!$. Substitute (3.4). The first three displayed terms use $P_1 = 1$, $P_2 = -(w + 2)$, and $P_3 = 2w^2 + 8w + 9$. $\square$

This form is often preferable to expanding at 0. If many values are needed near a known x_0 , one high-accuracy evaluation of w_0 supplies a local polynomial whose coefficients require only arithmetic.

12.2 Sensitivity to parameters

Suppose a model produces

$$w(\theta) e^{w(\theta)} = q(\theta). \quad (12.3)$$

Then $w(\theta) = W_k(q(\theta))$ on a fixed branch.

Proposition 12.2 (First parameter derivative). *If $q(\theta) \neq 0$ and $w(\theta) \neq -1$, then*

$$\frac{dw}{d\theta} = \frac{w}{q(1 + w)} \frac{dq}{d\theta}. \quad (12.4)$$

If $q(\theta_0) = 0$ on W_0 , the continuous limiting formula is $w'(\theta_0) = q'(\theta_0)$.

Proof. Apply the chain rule to $w = W_k(q)$ and use (3.1). At a principal-branch zero, $W'_0(0) = 1$ by theorem 3.1. $\square$

For a vector of parameters, the same formula gives the gradient

$$\nabla_{\theta} w = \frac{w}{q(1+w)} \nabla_{\theta} q.$$

The common factor exhibits the branch-point sensitivity independently of the application: all parameter derivatives inherit the factor $(1+w)^{-1}$.

12.3 A family of elementary antiderivatives

The substitution $x = we^w$ makes many integrals elementary, not only the ones in theorem 3.4. For $a \neq 0$ and an integer $n \geq 0$, define

$$E_n(a, w) = e^{aw} \sum_{j=0}^n \frac{(-1)^j n!}{(n-j)! a^{j+1}} w^{n-j}. \quad (12.5)$$

Lemma 12.3. *One has*

$$\frac{\partial}{\partial w} E_n(a, w) = w^n e^{aw}. \quad (12.6)$$

Proof. Differentiate (12.5). Multiplication of the sum by a produces w^n plus terms that cancel, term by term, against the derivative of the powers w^{n-j} . Explicitly, the derivative of the j th power term for $j < n$ is

$$e^{aw} \frac{(-1)^j n!}{(n-j-1)! a^{j+1}} w^{n-j-1},$$

which cancels the exponential derivative of the $(j+1)$ st term. Only the $j=0$ exponential derivative leaves $w^n e^{aw}$. $\square$

Theorem 12.4 (Polynomial moments of a branch). *Let m, r be nonnegative integers. On any differentiable branch interval,*

$$\int x^m W(x)^r dx = E_{m+r}(m+1, W(x)) + E_{m+r+1}(m+1, W(x)) + C. \quad (12.7)$$

Proof. Set $w = W(x)$. Then $x^m = w^m e^{mw}$ and $dx = e^w(1+w)dw$, so

$$\begin{aligned} \int x^m W(x)^r dx &= \int w^{m+r} (1+w) e^{(m+1)w} dw \\ &= \int (w^{m+r} + w^{m+r+1}) e^{(m+1)w} dw. \end{aligned}$$

Apply theorem 12.3 to the two terms. $\square$

Thus every integral of a nonnegative integer power of x times a nonnegative integer power of W has an elementary expression in $W(x)$ and $e^{W(x)}$. For noninteger powers the same substitution leads to incomplete gamma functions rather than finite polynomials.

12.4 Implicit differentiation without explicitly evaluating W

Sometimes one wants a derivative of the solution but not the solution's closed form. If w is defined by $we^w = x$, then

$$\frac{dx}{dw} = e^w(1 + w).$$

This gives, for any differentiable function H ,

$$\frac{d}{dx}H(W(x)) = \frac{H'(W(x))e^{-W(x)}}{1 + W(x)}. \quad (12.8)$$

The formula is often simpler than first differentiating a complicated expression in which W appears repeatedly.

Example 12.5. Let

$$Q(x) = \frac{W(x)^2}{1 + W(x)}.$$

Differentiating with respect to w first gives

$$\frac{d}{dw} \frac{w^2}{1 + w} = \frac{w(w + 2)}{(1 + w)^2}.$$

Therefore

$$Q'(x) = e^{-W(x)} \frac{W(x)(W(x) + 2)}{(1 + W(x))^3}.$$

This route avoids a longer quotient-rule calculation involving $W'(x)$.

12.5 Branch values as coordinates

The substitution

$$x = we^w \quad (12.9)$$

parameterizes each real branch without numerical inversion:

- $w \in [-1, \infty)$ traces the graph of W_0 ;
- $w \in (-\infty, -1]$ traces the graph of W_{-1} .

This is useful for plotting, quadrature, and testing inequalities. For example, a proposed principal-branch inequality $A(x) \leq W_0(x) \leq B(x)$ can be transformed into

$$A(we^w) \leq w \leq B(we^w), \quad w \geq -1,$$

which may be an ordinary one-variable inequality with no inverse function left in it.

13 Perspective and further directions

The Lambert W function sits at a productive boundary between elementary and special functions. Its defining map is elementary, yet its inverse has branches, a square-root critical point, logarithmic singularities, nontrivial asymptotics, and rich combinatorial coefficients.

Several themes recur throughout the article.

One geometry, two real branches. The principal branch and the lower branch are not unrelated formulas. They are the two inverse pieces of the same real curve. At $-1/e$ they share one signed Puiseux series; at their large-magnitude ends they share one logarithmic asymptotic expansion. Their differences—monotonicity, concavity, endpoint behavior, and branch selection—follow from which side of $w = -1$ is used.

Representations should be matched to regimes. The Taylor series is natural near 0 on W_0 , the signed- p series near the branch point, the logarithmic series for large $|W|$, and safeguarded iteration in the broad middle. No honest numerical implementation should expect one truncated formula to be uniformly excellent on both branches and all endpoints.

Bounds are algorithmic objects. The inequalities in section 7 are not only estimates. They create branch-safe starting intervals, prove monotone convergence, and separate intrinsic conditioning from algorithmic failure. Modern guaranteed evaluation methods refine this idea with interval arithmetic and carefully chosen recursions [15].

High-order inverse series remain an active tool. The general correction formula (8.11) turns any suitable approximation into a hierarchy of higher-order corrections. Recent work studies how to choose initial functions and organize these coefficients for both real branches [11]. Continued fractions, rational approximants, Chebyshev expansions, and verified complex arithmetic provide complementary approaches [2, 12, 4].

Generalized Lambert functions. Equations with several linear factors, such as

$$e^x \frac{(x - t_1) \cdots (x - t_p)}{(x - s_1) \cdots (x - s_q)} = a,$$

lead to generalized Lambert functions. Their branch geometry is richer, but the same strategy persists: analyze the forward map, locate critical values, construct local inverse series, and use logarithmic asymptotics at large solutions. The ordinary W function is the basic model on which that wider theory is built.

A practical final rule. When Lambert W appears in a calculation, always record the transformed variable’s admissible range before simplifying. The line

$$u = W_k(z)$$

is incomplete until the branch k is justified. On the real axis, most serious mistakes involving W are branch mistakes rather than algebra mistakes.

A Coefficient and branch tables

A.1 Taylor coefficients at the origin

The coefficient of x^n in $W_0(x)$ is $(-n)^{n-1}/n!$. The first twelve are listed for quick checking.

Table 3: First Taylor coefficients $[x^n]W_0(x)$.

n	coefficient	n	coefficient
1	1	7	16807/720
2	-1	8	-16384/315
3	3/2	9	531441/4480
4	-8/3	10	-156250/567
5	125/24	11	2357947691/3628800
6	-54/5	12	-2985984/1925

A.2 Puiseux coefficients

For

$$W = -1 + \sum_{n \geq 1} c_n p^n, \quad p = \pm \sqrt{2(1 + ex)},$$

the first coefficients are

Table 4: Coefficients shared by W_0 and W_{-1} at $-1/e$.

n	c_n	n	c_n
1	1	5	769/17280
2	-1/3	6	-221/8505
3	11/72	7	680863/43545600
4	-43/540	8	-1963/204120

A.3 Asymptotic correction polynomials

The first corrections in

$$W = L_1 - L_2 + \sum_{j \geq 1} \frac{P_j(L_2)}{L_1^j}$$

are summarized in table 5.

Table 5: First logarithmic asymptotic polynomials.

j	$P_j(L)$
1	L
2	$-L + \frac{1}{2}L^2$
3	$L - \frac{3}{2}L^2 + \frac{1}{3}L^3$
4	$-L + 3L^2 - \frac{11}{6}L^3 + \frac{1}{4}L^4$

The all-order formula is

$$P_j(L) = \sum_{m=1}^j \frac{(-1)^{j-m}}{m!} \left[\begin{matrix} j \\ j-m+1 \end{matrix} \right] L^m.$$

A.4 Real branch-selection checklist

Table 6: Choosing a real branch after reaching $ue^u = z$.

Information about trans- formed variable u	Argument z	Branch conclusion
$u \geq 0$	$z \geq 0$	only $W_0(z)$ is real
$-1 < u < 0$	$-1/e < z < 0$	$u = W_0(z)$
$u < -1$	$-1/e < z < 0$	$u = W_{-1}(z)$
$u = -1$	$z = -1/e$	$W_0 = W_{-1} = -1$
No range information	$-1/e < z < 0$	both real candidates must be checked in the original problem
Any real u	$z < -1/e$	no real solution

B A compact decision map for numerical evaluation

For a target precision of ordinary floating-point size, the following regime map is effective.

1. **Exact endpoints.** Return $W_0(0) = 0$ and $W_0(-1/e) = W_{-1}(-1/e) = -1$ exactly when the input representation permits.
2. **Near $-1/e$.** Compute $\eta = 1 + ex$ carefully and use the signed- p series. This avoids dividing by the small derivative of the forward map.
3. **Near 0 on W_0 .** Use the Taylor series or the $[2/2]$ Padé starter, followed by ordinary Newton or Halley.
4. **Large positive x .** Use $\log x - \log \log x$ or (9.18), followed by logarithmic Newton. Work with $\log x$ directly when x itself would overflow; this is the Wright omega formulation.
5. **Near 0^- on W_{-1} .** Use the logarithmic asymptotic expansion with $L_1 = \log(-x)$ and $L_2 = \log(-L_1)$, followed by logarithmic Newton.
6. **Guaranteed fallback.** Use the brackets in theorems 7.2 and 7.3 and the safeguarded algorithm 9.6.

A production implementation should also distinguish input error from iteration error. Near $-1/e$, the condition number in (3.6) can make a perfectly converged result contain few correct digits if the input itself was rounded too coarsely.

C Reproducing the computational supplement

The archive accompanying this article contains:

- `lambert_w_article.tex`, the complete source;
- `lambert_w_article.pdf`, the rendered article;
- `lambert_w_experiments.py`, the documented numerical script;

- the generated PDF figures in `figures/`;
- sampled numerical results in `data/numerical_results.csv`;
- `numerical_results.tex`, generated by the script and included in the article.

From a Python environment containing `mpmath`, `numpy`, and `matplotlib`, run

```
python lambert_w_experiments.py --output-dir .
```

Then compile with a modern LaTeX installation, for example

```
latexmk -pdf -interaction=nonstopmode -halt-on-error lambert_w_article.tex
```

The script uses 80 decimal digits for reference calculations. Plotting arrays are converted to ordinary floating point only after the high-precision values have been computed. Every approximation in the script is also written as a separate Python function, making it straightforward to change grids or add a new candidate formula.

D Bibliographic notes on the supplied source archive

The foundational modern reference is Corless et al. [5], which systematized the branch notation, complex asymptotics, symbolic integration, and numerical evaluation. The DLMF entry [7] gives a concise reference for standard branch conventions and expansions.

The numerical and approximation literature represented in the supplied archive includes the piecewise real approximations of Barry et al. [2], the global principal-branch approximations and logarithmic Newton iteration of Iacono and Boyd [12], the guaranteed high-precision analysis of Lóczi [15], and Howard’s Schröder-type expansions [11]. The lower-branch enclosure proved in theorem 7.3 is based on the inequalities of Chatzigeorgiou [3]. Corcino, Corcino, and Mező [4] study integral and continued-fraction representations.

The application papers in the archive illustrate why explicit branch-aware forms matter: Banwell treats transistor and diode circuits [1]; Goličnik treats biochemical kinetics [10]; Lehtonen surveys ecological and evolutionary models [14]; and Visser discusses prime-number approximations [17]. These applications motivated several examples in section 11, but all derivations in this article have been carried out independently.

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